



安徽理工大学
ANHUI UNIVERSITY OF SCIENCE & TECHNOLOGY

计算机科学与工程学院
本科毕业设计（论文）附件材料

中文题目：基于深度学习的车道障碍检测与碰撞预测

方法研究

专 业：计算机科学与技术

班 级：计算机 21-4 班

学 号：2021304369

姓 名：陈亚辉

指导教师：郑苹 职称讲师

企业导师：杨显波 职称工程师

2025 年 6 月

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文献翻译

学院（部）： 计算机科学与工程学院
班 级： 计算机 21-4 班
学生姓名： 陈亚辉
指导教师： 郑苹 讲师

2025 年 6 月

安徽理工大学毕业论文（设计）外文文献翻译评分表

文献来源： 期刊论文 ☒ 会议论文 ☐ 专著 ☐ 其他：__ ☐

翻译字数：30000（原文）

评分维度	评分标准	分值	得分
1. 专业匹配性	- 选择合理，与论文（设计）内容相关性高（16-20分）	20	18
	- 选择基本合理，与论文（设计）内容具有一定相关性（11-15分）		
	- 选择不合理，与论文（设计）内容相关性差（0-10分）		
2. 翻译准确性	- 忠实原文，无漏译、误译（16-20分）	20	18
	- 基本准确，偶有小错误（11-15分）		
	- 有明显错误或偏差（6-10分）		
3. 语言流畅性	- 严重偏离原文（0-5分）	20	17
	- 中文表达通顺，符合学术规范（16-20分）		
	- 基本通顺，个别语句生硬（11-15分）		
4. 专业术语规范	- 多处语病或口语化（6-10分）	20	17
	- 语句混乱难懂（0-5分）		
	- 术语翻译准确且统一（16-20分）		
5. 格式与排版	- 术语基本正确，个别不统一（11-15分）	20	17
	- 术语错误较多（6-10分）		
	- 术语混乱（0-5分）		
6. 翻译态度与进度	- 符合学校格式要求，标注清晰（10分）	10	8
	- 基本符合，少量格式问题（7-9分）		
	- 格式混乱（0-6分）		
7. 翻译态度与进度	- 按时提交，提交前自主校对（10分）	10	8
	- 延期≤3天，校对不足（7-9分）		
	- 延期严重或未校对（0-6分）		

总得分：86分

评语：（围绕主要优点和不足之处填写）

符合本科毕业答辩要求。

评分教师：郑苹

日期：2025年05月28日

Despite the recent advancements that Autonomous Vehicles have shown in their potential to improve safety and operation, considering differences between Autonomous Vehicles and Human-Driven Vehicles in accidents remain unidentified due to the scarcity of real-world Autonomous Vehicles accident data. We investigated the difference in accident occurrence between Autonomous Vehicles' levels and Human-Driven Vehicles by utilizing 2100 Advanced Driving Systems and Advanced Driver Assistance Systems and 35,113 Human-Driven Vehicles accident data. A matched case-control design was conducted to investigate the differential characteristics involving Autonomous' versus Human-Driven Vehicles' accidents. The analysis suggests that accidents of vehicles equipped with Advanced Driving Systems generally have a lower chance of occurring than Human-Driven Vehicles in most of the similar accident scenarios. However, accidents involving Advanced Driving Systems occur more frequently than Human-Driven Vehicle accidents under dawn/ dusk or turning conditions, which is 5.25 and 1.98 times higher, respectively. Our research reveals the accident risk disparities between Autonomous Vehicles and Human-Driven Vehicles, informing future development in Autonomous technology and safety enhancements.

尽管自动驾驶汽车在提升安全性和操作性能方面已经取得了一些进展，但因为现实世界中自动驾驶汽车的事故数据比较少，所以自动驾驶汽车和人类驾驶汽车在事故中的区别还是很难确定。我们用 2100 个高级驾驶系统和高级驾驶辅助系统，还有 35113 个人类驾驶汽车事故数据，研究了自动驾驶汽车和人类驾驶汽车在事故发生率上的不同。我们用匹配病例对照设计来分析自动驾驶汽车和人类驾驶汽车事故的差异。分析发现，在大多数类似的事故情况下，用高级驾驶系统的汽车发生事故的概率通常比人类驾驶汽车低。但是，在黎明或黄昏时段，或者转弯的时候，用高级驾驶系统的汽车发生事故的频率是比人类驾驶汽车高 5.25 倍和 1.98 倍。我们的研究揭示了自动驾驶汽车和人类驾驶汽车之间的事故风险差异，为自动驾驶技术和安全提升的未来发展提供了参考。

Findings of road, environment, and accident type Based on the results of the matched case control logistic regression model, compared with HDV, the odds of an ADS accident occurring in rainy weather are 0.335 times. This indicates a lower likelihood of an ADS accident in rainy weather compared to an HDV accident. RADAR is capable of detecting objects at distances exceeding 150 m, even in adverse weather conditions such as fog or rain³⁴. In contrast, human drivers may only be able to

perceive objects up to approximately 10 meters away under similar circumstances³⁵. Although adverse weather can increase the likelihood of potential failures or loss of sensors^{36–38}, recent innovations in visual algorithms, coupled with the combined use of cameras, LIDAR, GNSS, and RADAR sensors^{39,40}, are crafted to recognize pedestrians and vehicles under varying weather scenarios, such as cloudiness, snow, rain, and darkness^{41,42}. This offers solutions to the challenges associated with driving in less-than-ideal conditions. In contrast, human drivers may have difficulties seeing through heavy rain or fog, leading to a delay in detecting potential hazards or reacting appropriately⁴⁰. Interestingly, the dawn/dusk odds ratio indicates a 5.250 higher probability of ADS accident than HDV accident. This could be attributed to the sensors and cameras used by AVs may not be able to quickly adapt to changes in lighting conditions, which could affect their ability to detect obstacles, pedestrians, and other vehicles^{39,43}. At dawn and dusk, for instance, the sun's shadows and reflections may confuse sensors, making it hard for them to distinguish between objects and identify potential hazards. Furthermore, the fluctuating light conditions can impact the accuracy of object detection and recognition algorithms used by AVs, which can result in false positives or negatives^{35,44}. Accident types related findings for ADS and HDVs are worth noting. Compared to HDV accidents, AVs experience relatively lower risks in rear-end and broadside accidents (0.457 times and 0.171 times, respectively). This finding indicates that AVs may detect and react to potential rear-end and sideswipe accident situations much faster than humans can. This is because they are equipped with advanced sensors and software that can quickly analyze the surrounding environment and make decisions based on the data received⁴⁵. In addition, the kinematic method used by ACC system keeps track of and regulates the distance between vehicles, alerting drivers if this space becomes smaller than the safe limit, especially at highways⁴⁶. By ensuring that vehicles keep a consistent speed and distance between vehicles, thus effectively reduces the risk of rear-end accidents⁴⁷. Compared with ADS, HDVs tend to display greater velocity differences at larger spacing ranges⁴⁸, a factor that significantly contributes to a higher incidence of rear-end and sideswipe accidents⁴⁹. Findings of pre-accident conditions and accident outcomes In terms of pre-accident conditions, most of the pre-accident movements made by ADS reduce the probability of accidents from the results of the matched case control logistic regression, except for turning, which increases the likelihood of an accident by 1.988 times compared to HDVs. One

possible reason is a lack of situational awareness. Situational awareness of AVs can be defined as the ability of these vehicles to perceive essential elements in their surroundings understand the importance of these elements, and anticipate their future state or changes⁵⁰. The complexity of turning in autonomous driving scenarios arises from three primary challenges: choosing the appropriate lane (target lane selection), devising and computing a safe and efficient path (trajectory planning and calculation), and executing the turn while adjusting to dynamic conditions (vehicle controlling and tracking)⁵¹. AVs rely on sensors and algorithms to perceive their surroundings and make driving decisions⁴⁵. However, these systems may not detect all obstacles and hazards, particularly in complex and dynamic driving scenarios like turning at intersections^{52,53}. It is a significant challenge to generate sufficient information and achieve comprehensive detection of the surrounding environment from a single independent source due to limited sensor ranges and limited coverage of the environment by sensors in AV^{45,54}. Additionally, some AVs are programmed to follow predefined rules and scenarios, which may not encompass every possible driving situation^{55–57}. The modifications of scenarios can present difficulty for AVs in perceiving and responding to them, thereby raising the risk of an accident⁵⁸. Moreover, multiple oncoming HDVs and the complexity of such driving scenarios are a considerable challenge for AVs such as unprotected left turns at intersections⁵⁹. These situations are complicated by factors like limited priority and variation in trajectories⁶⁰. AVs tend to be overcautious (such as having a longer startup delay during the turning at intersections)⁶¹, which can lead to rear-end or sideswipe accidents with HDVs⁶². Furthermore, multi-interactions caused by mixed flows aggravate uncertainties in detection, such as the superposition of distance and angle measurement error⁶³. Conversely, HDVs can adapt and modify their speed more seamlessly than AVs, highlighting the limitations of AVs in comparison to the adeptness of experienced drivers⁶⁴. And AVs face difficulties with executing lane changes or turning in heavy traffic and lack psychological insight^{65,66}. In addition, HDVs can predict pedestrian movements and exercise caution based on their driving experience, whereas AVs may struggle with recognizing pedestrians' intentions, potentially leading to emergency braking or accidents due to a lack of understanding of social cues and psychological reasoning^{65,67,68}. ADS accidents are less likely to occur than HDV accidents in situations such as proceeding straight, run-off road (a vehicle leaves the designated roadway and travels onto an area that is not intended for

regular traffic) and entering traffic lane conditions (a vehicle transitioning from a stationary or parked position to enter a traffic lane and become physically present within the flow of traffic). When considering the proceeding straight condition, it was found that AV accident resulted in a 0.299 lower probability of an HDV accident. Remarkably, ADS accident risk is 0.021 times as high as that of an HDV accident in run-off road condition, which can be explained by the faster reaction time of AVs²⁴. AVs can detect these situations and apply corrective actions, such as adjusting the speed or steering angle^{69,70}, more quickly and accurately than a human driver^{71,72}. The result of matched-case control model revealed a significant correlation between the entering traffic lane condition of ADS accident, the risk of which is 0.267 times as high as HDV accident. According to the results of the matched casecontrol logistic regression, the impact of backing is noteworthy, which shows that the ADS is less likely to be affected than the HDV. According to the analysis, the model using accidents of AVs showed a decreased probability of accidents for moderate and fatal severity in comparison to HDV.

道路、环境和事故类型的发现，根据病例对照逻辑回归模型的结果，与重型货车相比，在雨天发生 ADS 事故的概率是后者的 0.335 倍。这说明，与重型货车事故相比，在雨天发生 ADS 事故的可能性较低。即使在雾或雨等恶劣天气条件下，雷达也能探测到 150 米以外的物体。相比之下，人类驾驶员在类似情况下可能只能感知到约 10 米外的物体。尽管恶劣天气会增加传感器潜在故障或丢失的可能性，但近年来视觉算法的创新，加上摄像头、激光雷达、GNSS 和雷达传感器的结合使用，可以在阴天、雪天、雨天和黑暗等各种天气情景下识别行人和车辆。这为在非理想条件下驾驶带来的挑战提供了解决方案。相比之下，人类驾驶员可能难以看清大雨或大雾，从而导致无法及时发现潜在危险或做出适当反应。有趣的是，黎明/黄昏的概率表明，自动驾驶汽车（ADS）事故的概率比重型货车（HDV）事故高出 5.250 倍。这可能是因为自动驾驶汽车使用的传感器和摄像头可能无法快速适应光照条件的变化，从而影响其检测障碍物、行人和其他车辆的能力。例如，在黎明和黄昏时分，太阳的阴影和反射可能会干扰传感器，使其难以区分物体并识别潜在危险。此外，光照条件的波动也会影响自动驾驶汽车使用的物体检测和识别算法的准确性，从而导致误报或漏报。值得关注的是，自动驾驶汽车（ADS）和重型货车（HDV）的事故类型相关发现。与重型货车（HDV）事故相比，自动驾驶汽车（AV）发生追尾和侧撞事故的风险相对较低（分别为 0.457 倍和 0.171 倍）。这一发现表明，自动驾驶汽车能够比人类更快地发现并应对潜在的追尾和侧撞事故情况。这是因为

它们配备了先进的传感器和软件，可以快速分析周围环境并根据接收到的数据做出决策。此外，自适应巡航（ACC）系统采用的运动学方法会跟踪和调节车辆之间的距离，如果该距离小于安全限值，尤其是在高速公路上，则会向驾驶员发出警报。通过确保车辆保持一致的速度和车距，可以有效降低追尾事故的风险。与 ADS 相比，重型货车（HDV）在更大的车距范围内往往表现出更大的速度差异，这一因素显著导致追尾和侧撞事故的发生率更高。理解这些要素的重要性，并预测它们未来的状态或变化。自动驾驶场景中转弯的复杂性源于三个主要挑战：选择合适的车道（目标车道选择）、设计和计算安全有效的路径（轨迹规划和计算）以及在根据动态条件调整的同时执行转弯（车辆控制和跟踪）。自动驾驶汽车依靠传感器和算法来感知周围环境并做出驾驶决策。然而，这些系统可能无法检测到所有障碍物和危险，特别是在复杂和动态的驾驶场景中，例如在十字路口转弯。由于传感器范围有限以及自动驾驶汽车中传感器对环境的覆盖有限，从单一独立来源生成足够的信息并实现对周围环境的全面检测是一项重大挑战。此外，一些自动驾驶汽车被编程为遵循预定义的规则和场景，这些规则和场景可能无法涵盖所有可能的驾驶情况。场景的变化会使 AV 难以感知和响应，从而增加发生事故的风险。此外，多辆迎面驶来的 HDV 以及此类驾驶场景的复杂性对 AV 来说是一个巨大的挑战，例如在交叉路口无保护的左转。这些情况由于优先级有限和轨迹变化等因素而变得复杂。AV 往往过于谨慎（例如在交叉路口转弯时启动延迟较长），这可能导致与 HDV 发生追尾或侧撞事故。此外，混合流引起的多重交互加剧了检测的不确定性，例如距离和角度测量误差的叠加。相反，HDV 可以比 AV 更无缝地调整和修改其速度，突显了 AV 与经验丰富的驾驶员的熟练程度相比的局限性。并且，AV 在交通拥堵时难以执行车道变换或转弯，并且缺乏心理洞察力。此外，重型货车可以预测行人的运动并根据驾驶经验谨慎驾驶，而自动驾驶汽车可能难以识别行人的意图，由于缺乏对社交线索和心理推理的理解，可能导致紧急制动或事故。在直线、驶离道路（车辆离开指定道路并驶入非正常交通区域）和进入车道条件（车辆从静止或停放位置过渡到车道并实际出现在车流中）等情况下，ADS 事故发生的可能性低于 HDV 事故。当考虑直线条件时，研究发现 AV 事故导致 HDV 事故的概率降低 0.299。值得注意的是，在驶离道路条件下，ADS 事故风险是 HDV 事故的 0.021 倍，这可以通过 AV 的反应时间更快来解释。AV 可以检测到这些情况并采取纠正措施，例如比人类驾驶员更快、更准确地调整速度或转向角度。病例对照模型结果显示，ADS 事故的入口条件与入口条件存在显著相关性，其风险是 HDV 事故的 0.267 倍。病例对照逻辑回归结果显示，倒置效应显著，表明 ADS 受影响的可能性小于 HDV。分析表

明，使用自动驾驶汽车事故的模型显示，与 HDV 相比，中度和致命事故的概率有所降低。

Waymo's mission to reduce traffic injuries and fatalities and improve mobility for all has led us to expand deployment of automated vehicles (AVs) on public roads without a human driver behind the wheel. As part of this process, Waymo is committed to providing the public with informative and relevant data regarding the demonstrated safety of Waymo's automated driving system (ADS), which we call the Waymo Driver. The data presented in this paper represents more than 6.1 million miles of automated driving in the Phoenix, Arizona metropolitan area, including operations with a trained operator behind the steering wheel from calendar year 2019 and 65,000 miles of driverless operation without a human behind the steering wheel from 2019 and the first nine months of 2020. The paper includes every collision and minor contact experienced during these operations as well as every predicted contact identified using Waymo's counterfactual ("what if") simulation of events had the vehicle's trained operator not disengaged automated driving. There were 47 contact events that occurred over this time period, consisting of 18 actual and 29 simulated contact events, none of which would be expected to result in severe or life-threatening injuries. This paper presents the collision typology and severity for each actual and simulated event, along with diagrams depicting each of the most significant events. Nearly all the events involved one or more road rule violations or other errors by a human driver or road user, including all eight of the most severe events (which we define as involving actual or expected airbag deployment in any involved vehicle). When compared to national collision statistics, the Waymo Driver completely avoided certain collision modes that human-driven vehicles are frequently involved in, including road departure and collisions with fixed objects. While data related to these collision modes is very promising, the presence of collisions that resulted from challenging situations induced by other drivers serves as a reminder of the limits of AV collision avoidance as long as AVs share roadways with human drivers. Analysis of events from AV operation on public roads is only one of many complementary safety evaluation methods that Waymo uses, and we are sharing it because it is objective and directly relevant to public road operation of AVs. As automated vehicles continue to improve and fleet mileage continues to grow, so will public understanding of their safety impact. The long-term contributions of this paper are not

only the events and mileages shared, but the example set by publicly sharing this type of safety information.

Waymo 的目标就是减少交通事故，让大家都能更方便地出行。所以我们不仅在公共道路上推广无人驾驶汽车，还努力向公众展示我们的自动驾驶系统（也就是我们说的 Waymo Driver）的安全性。这篇文章里，我们分享了在亚利桑那州凤凰城地区超过 610 万英里的自动驾驶行驶数据，包括从 2019 年开始的由专业驾驶员驾驶的运营情况，还有 2019 年到 2020 年前九个月 65,000 英里的无人驾驶运营情况。我们记录了所有碰撞和轻微接触事件，还有通过模拟识别出的预测接触。在这段时间里，发生了 47 起接触事件，包括 18 起实际接触和 29 起模拟接触，预计这些事件都不会造成严重伤害。我们介绍了每起事件的具体情况和严重程度，还附上了图表。几乎所有事件都是因为人类驾驶员或道路使用者违反了交通规则或其他失误造成的，包括所有八起最严重的事件（我们定义为可能触发安全气囊的事件）。与全国碰撞数据相比，Waymo Driver 完全避免了人类驾驶车辆常见的碰撞模式，比如偏离道路和与固定物体的碰撞。尽管这些数据很有价值，但其他驾驶员引起的复杂情况导致的碰撞提醒我们，只要自动驾驶汽车和人类驾驶员共用道路，自动驾驶汽车的防撞能力就有局限性。我们对自动驾驶汽车在公共道路上的运行事件进行分析，只是我们众多安全评估方法之一。我们分享这些信息，是因为它们客观且与自动驾驶汽车在公共道路上的运行直接相关。随着自动驾驶汽车不断改进和行驶里程的增加，公众对其安全性的理解也会不断加深。这篇文章的长期贡献不仅在于分享的事件和行驶里程，还在于公开分享此类安全信息所树立的榜样。

This paper provides a detailed analysis of every actual and simulated, counterfactual (“what if”) collision or contact that was collected from more than 6.1 million miles of fully automated driving in Waymo’s metro Phoenix ODD. This includes operations with a trained operator behind the steering wheel from calendar year 2019 and 65,000 miles of driverless operation without a human behind the steering wheel from 2019 and the first nine months of 2020. Taken together, these 47 lower severity (S0 and S1) events (18 actual and 29 simulated, one during driverless operation) show significant contribution from other agents, namely human-related deviations from traffic rules and safe driving performance. Nearly all the actual and simulated events involved one or more road rule violations or other incautious behavior by another agent, including all eight of the most severe events involving actual or expected airbag deployment. The frequency of challenging events that were induced by incautious behaviors of

other drivers serves as a clear reminder of the challenges in collision avoidance so long as AVs share roadways with human drivers. Statistics regarding the high percentage of human collisions that are attributed to human error may lead to inflated expectations of the potential immediate safety benefits of AVs. AVs will share roads with human drivers for the foreseeable future, and significant numbers of collisions due to human driver errors that are simply unavoidable should be expected during this period. Due to the typology of those collisions initiated by other actors as well as the Waymo Driver's proficiency in avoiding certain collision modes, the data presented shows a significant shift in the relative distributions of collision types as compared to national crash statistics for human drivers. For example, the Waymo Driver experienced zero actual or simulated collision-relevant contacts in the NHTSA "road departure, fixed object, rollover" single-vehicle collision typology (27% of all US roadway fatalities). In an additional example, while rear-end collisions are one of the most common collision modes for human drivers, the Waymo Driver only recorded a single front-to-rear striking collision (simulated) and that event involved an agent cutting in and immediately braking (consistent with antagonistic motive). This is the first time that information on every actual and simulated collision or contact has been shared for millions of miles of automated driving. Data from more than 6.1 million miles of driving (representing over 500 years of driving for the average U.S. licensed driver) provides sufficient statistical signal to detect moderate-to-large differences in S0 and S1 event frequencies. However, as discussed in Waymo's Safety Methodologies and Safety Readiness Determinations 1 our assessment of AV safety uses multiple complementary methodologies, including simulation and closed course testing, which allows for comprehensive testing, including rare scenarios beyond those encountered in this dataset. As AV fleets and their mileage continue to grow, so will our understanding of their safety impact. This data only represents the performance of the Waymo Driver at a snapshot in time, and the performance of the Waymo Driver is continually improving. Therefore, the most significant long-term contributions of this paper will likely not be the actual data shared, but the example set by publicly sharing this type of safety performance data and the dialogs that this paper fosters.

本文对 Waymo 凤凰城市区超过 610 万英里的全自动驾驶中收集到的每一次实际、模拟、反事实（“假设”）碰撞或接触进行了详细分析。这包括自 2019 年以来由训练有素的驾驶员驾驶的运营情况，以及自 2019 年起至 2020 年

前九个月，无人驾驶情况下的 65,000 英里无人驾驶运营情况。综合起来，这 47 起低严重程度（S0 和 S1）事件（18 起实际事件、29 起模拟事件、1 起无人驾驶期间事件）显示出其他因素的显著影响，即与人为因素相关的交通规则偏差和安全驾驶表现。几乎所有实际和模拟事件都涉及另一名代理人的某项或多项违反道路规则或其他不谨慎行为，包括涉及实际或预期安全气囊展开的所有 8 起最严重事件。由于其他驾驶员的不谨慎行为而引发的碰撞事故频发，这清楚地表明，只要自动驾驶汽车与人类驾驶员共享道路，碰撞规避就将面临诸多挑战。统计数据表明，人为失误导致的人为碰撞事故比例很高，这可能会导致人们对自动驾驶汽车潜在的即时安全效益抱有过高的期望。在可预见的未来，自动驾驶汽车仍将与人类驾驶员共享道路，在此期间，预计会有大量因不可避免的人为失误而发生的碰撞事故。由于其他参与者引发的碰撞类型以及 Waymo 驾驶员在避免某些碰撞模式方面的熟练程度，与人类驾驶员的全国碰撞统计数据相比，所呈现的数据显示碰撞类型的相对分布发生了显著变化。例如，在 NHTSA “道路偏离、固定物体、翻车” 单车碰撞类型（占美国所有道路死亡人数的 27%）中，Waymo 驾驶员没有经历过任何实际或模拟的碰撞相关接触。再举一个例子，虽然追尾碰撞是人类驾驶员最常见的碰撞模式之一，但 Waymo 驾驶员只记录了一次前后撞击（模拟），该事件涉及一名代理切入并立即刹车（与对抗动机一致）。这是首次在数百万英里的自动驾驶里程中共享所有实际和模拟碰撞或接触信息。超过 610 万英里的驾驶数据（相当于美国持照驾驶员平均 500 多年的驾驶经验）提供了足够的统计信号，可以检测到 S0 和 S1 事件频率的中度到较大差异。然而，正如 Waymo 的安全方法和安全准备度判定中所讨论的，自动驾驶汽车安全评估采用了多种互补的方法，包括模拟和封闭式赛道测试，这使得测试能够涵盖本数据集中未曾遇到的罕见场景。随着自动驾驶车队及其行驶里程的不断增长，我们对其安全影响的理解也将不断加深。这些数据仅代表 Waymo Driver 在某一特定时间点的表现，而 Waymo Driver 的表现正在不断提升。因此，本文最重要的长期贡献可能不是共享的实际数据，而是通过公开分享此类安全性能数据所树立的榜样以及本文所促进的对话。

The semantic image segmentation task consists of classifying each pixel of an image into an instance, where each instance corresponds to a class. This task is a part of the concept of scene understanding or better explaining the global context of an image. In the medical image analysis domain, image segmentation can be used for image-guided interventions, radiotherapy, or improved radiological diagnostics. In this

review, we categorize the leading deep learning-based medical and non-medical image segmentation solutions into six main groups of deep architectural, data synthesis-based, loss function-based, sequenced models, weakly supervised, and multi-task methods and provide a comprehensive review of the contributions in each of these groups. Further, for each group, we analyze each variant of these groups and discuss the limitations of the current approaches and present potential future research directions for semantic image segmentation.

语义图像分割的任务就是给图像里的每个像素都分个类，每个类代表一个实例。这就像是理解场景的一部分，或者说是图像的整体背景有个更清楚的解释。在医学图像分析这块，图像分割能帮助做图像引导手术、放射治疗，或者提高放射诊断的准确性。在这篇综述里，我们把基于深度学习的医学和非医学图像分割方法分成六大类：深度架构、数据合成、损失函数、序列模型、弱监督和多任务方法，对每种方法的贡献都做了全面梳理。而且，对于每一类方法，我们还分析了它们的各种变体，讨论了现有方法的不足，还提出了语义图像分割未来可能的研究方向。

Guo et al. (2018) provided a review of deep learning based semantic segmentation of images, and divided the literature into three categories: region-based, fully convolutional network (FCN)-based, and weakly supervised segmentation methods. Hu et al. (2018b) summarized the most commonly used RGB-D datasets for semantic segmentation as well as traditional machine learning based methods and deep learning-based network architectures for RGB-D segmentation. Lateef and Ruichek (2019) presented an extensive survey of deep learning architectures, datasets, and evaluation methods for the semantic segmentation of natural images using deep neural networks. Similarly, for medical imaging, Goceri and Goceri (2017) presented an high-level overview of deep learning-based medical image analysis techniques and application areas. Hesamian et al. (2019) presented an overview of the state-of-the-art methods in medical image segmentation using deep learning by covering the literature related to network structures and model training techniques. Karimi et al. (2019) reviewed the literature on techniques to handle label noise in deep learning based medical image analysis and evaluated existing approaches on three medical imaging datasets for segmentation and classification tasks. Zhou et al. (2019b) presented a review of techniques proposed for fusion of medical images from multiple modalities for medical image segmentation. Goceri (2019a) discussed the fully supervised,

weakly supervised and transfer learning techniques for training deep neural networks for segmentation of medical images, and also discussed the existing methods for addressing the problems of lack of data and class imbalance. Zhang et al. (2019) presented a review of the approaches to address the problem of small sample sizes in medical image analysis, and divided the literature into five categories including explanation, weakly supervised, transfer learning, and active learning techniques. Tajbakhsh et al. (2020) presented a review of the literature for addressing the challenges of scarce annotations as well as weak annotations (e.g., noisy annotations, image-level labels, sparse annotations, etc.) in medical image segmentation. Similarly, there are several surveys covering the literature on the task of object detection (Wang et al., 2019c; Zou et al., 2019; Borji et al., 2019; Liu et al., 2019b; Zhao et al., 2019), which can also be used to obtain what can be termed as rough localizations of the object(s) of interest. In contrast to the existing surveys, we make the following contributions in this review: – We provide comprehensive coverage of research contributions in the field of semantic segmentation of natural and medical images. In terms of medical imaging modalities, we cover the literature pertaining to both 2D (RGB and grayscale) as well as volumetric medical images. – We group the semantic image segmentation literature into six different categories based on the nature of their contributions: architectural improvements, optimization function based improvements, data synthesis based improvements, weakly supervised models, sequenced models, and multi-task models. Figure 1 indicates the categories we cover in this review, along with a timeline of the most influential papers in the respective categories. Moreover, Figure 2 shows a high-level overview of the deep semantic segmentation pipeline, and where each of the categories mentioned in Figure 1 belong in the pipeline. – We study the behaviour of many popular loss functions used to train segmentation models on handling scenarios with varying levels of false positive and negative predictions. – Followed by the comprehensive review, we recognize and suggest the important research directions for each of the categories.

Guo 等人 (2018) 回顾了基于深度学习的图像语义分割, 把文献分成了三类: 基于区域的、基于全卷积网络 (FCN) 的和弱监督的分割方法。Hu 等人 (2018b) 总结了最常用的用于语义分割的 RGB-D 数据集, 以及用于 RGBD 分割的传统机器学习方法和基于深度学习的网络架构。Lateef 和 Ruichek (2019) 对使用深度神经网络进行自然图像语义分割的深度学习架构、数据集和评估方法进行了广泛的综述。同样, 对于医学成像, Goceri 和 Goceri

(2017) 对基于深度学习的医学图像分析技术和应用领域进行了高级概述。Hesamian 等人 (2019) 通过涵盖与网络结构和模型训练技术相关的文献, 概述了使用深度学习进行医学图像分割的最新方法。Karimi 等人 (2019) 回顾了基于深度学习的医学图像分析中处理标签噪声技术的文献, 并评估了在三个医学成像数据集上用于分割和分类任务的现有方法。Zhou 等人 (2019b) 回顾了用于融合多模态医学图像进行医学图像分割的技术。Goceri (2019a) 讨论了用于训练深度神经网络进行医学图像分割的全监督、弱监督和迁移学习技术, 并讨论了现有的解决数据缺乏和类别不平衡问题的方法。Zhang 等人 (2019) 回顾了解决医学图像分析中样本量较小的问题的方法, 并将文献分为五类, 包括解释、弱监督、迁移学习和主动学习技术。Tajbakhsh 等人 (2020) 对医学图像分割中应对稀缺注释和弱注释 (例如噪声注释、图像级标签、稀疏注释等) 挑战的文献进行了综述。同样, 也有几篇涵盖目标检测任务文献的综述 (Wang 等人, 2019c; Zou 等人, 2019; Borji 等人, 2019; Liu 等人, 2019b; Zhao 等人, 2019), 这些综述也可用于获取感兴趣对象的粗略定位。与现有综述相比, 我们在本综述中做出了以下贡献: 我们全面涵盖了自然图像和医学图像语义分割领域的研究成果。在医学成像模式方面, 我们涵盖了二维 (RGB 和灰度) 以及体积医学图像的相关文献。我们根据语义图像分割文献的贡献性质, 将其分为六类: 架构改进、基于优化函数的改进、基于数据合成的改进、弱监督模型、序列模型和多任务模型。图 1 列出了本综述中涵盖的类别, 以及各领域最具影响力的论文的时间表。我们研究了许多流行的损失函数的行为, 这些函数用于训练分割模型来处理具有不同程度假阳性和假阴性预测的场景。经过全面审查, 我们认识到并建议重要的研究管道。类别。此外, 图 2 展示了深度语义分割流程的高级概述, 以及图 1 中提到的每个类别在每个方向。

Knowledge distillation transfers the knowledge from a cumbersome teacher to a small student. Recent results suggest that the student-friendly teacher is more appropriate to distill since it provides more transferable knowledge. In this work, we propose the novel framework, “prune, then distill,” that prunes the model first to make it more transferrable and then distill it to the student. We provide several exploratory examples where the pruned teacher teaches better than the original unpruned networks. We further show theoretically that the pruned

teacher plays the role of regularizer in distillation, which reduces the generalization error. Based on this result, we propose a novel neural network compression scheme where the student network is formed based on the pruned teacher and then apply the “prune, then distill” strategy.

知识蒸馏将知识从笨重的“老师”网络转移到小学生网络。最近的结果表明，学生友好的“老师”网络更适合蒸馏，因为它提供了更多可迁移的知识。在本研究中，我们提出了一个新颖的框架——“先剪枝，后蒸馏”，该框架首先对模型进行剪枝，使其更具可迁移性，然后将其蒸馏给学生网络。我们提供了几个探索性示例，其中剪枝后的“老师”网络比原始未剪枝网络的教学效果更好。我们进一步从理论上证明，剪枝后的“老师”网络在蒸馏过程中扮演着正则化器的角色，从而降低了泛化误差。基于此结果，我们提出了一种新颖的神经网络压缩方案，其中基于剪枝后的“老师”网络构建学生网络，然后应用“先剪枝，后蒸馏”策略。

We conduct experiments to verify the effectiveness of pruned teachers in KD. Instead of distilling the teacher network directly (dotted-blue block in Figure 1), we first (unstructured) prune the teacher network and then distill to the student network (dotted-red block in Figure 1). Setups: We mainly consider VGG [43] and ResNet 16 for the teacher network, where VGG is trained on the CIFAR100 dataset [24] and ResNet is trained on the TinyImageNet dataset [25]. The TinyImageNet dataset is a subset of resized ($3 \times 64 \times 64$) ImageNet dataset 6. We reserve 10% of the data as a validation set in all training. We apply unstructured pruning that removes more weights, more precisely LR rewinding [39], to prune the teacher model. In LR rewinding, we set the ratio of epochs by 0.65 for VGG-CIFAR100. In other words, we train the VGG19 for 200 epochs initially, then rewind the learning rates and retrain (fine-tuning) the network for 130 epochs (65%). Note that the different ratios from 0.6 to 0.9 do not make significant differences in pruning, and we use the ratio of 0.5 for ResNet-TinyImageNet. For a fair comparison, we train (and distill) networks with enough epochs and halt the training at their best performance on validation dataset. All test accuracies are the average of three independent experiments, and we also provide the standard deviation. For simplicity, we use the vanilla KD 20 with appropriate balancing parameter α and temperature τ . The balancing parameter α represents the ratio of two objectives (distill loss and hard-target loss). The

temperature τ is a softening parameter, where higher τ produces a softer target. In the experiment, we fix the parameters by $\alpha = 0.95$ and $\tau = 10$. More detailed training parameters are provided in Appendix. Note that the purpose of experiments is not achieving the best test accuracy but to compare between distilling from a pruned network and distilling from an unpruned network. Thus, hyperparameters, as well as network architectures, are not optimized for test accuracies. Instead, we use as-is settings for a fair comparison between the pruned teacher and the unpruned teacher. For example, we follow default settings for MobileNetV2 [42] and ResNet18 16 optimized for ImageNet dataset 6, while we use TinyImageNet dataset [25].

Distill VGG19 to VGG11: We set VGG19 [43] as a teacher network and VGG11 as a student network. The network architecture of VGG is unchanged except for the number of fully connected (FC) layers, where our VGG has a single FC layer (which is commonly used for CIFAR10 data). Then, we compare the KD results on the CIFAR100 dataset [24] between the regular VGG19 teacher and the pruned VGG19 teacher. We prune the teacher network with three sparsity levels: 36% sparsity (36% of weights are removed), 59% sparsity, and 79% sparsity. Surprisingly, as shown in Table 1, VGG11 with pruned VGG19 consistently outperforms the one with the unpruned teacher. Table 1 also provides results when the teacher network is VGG19DBL, with $2 \times$ many channels in each layer. In both cases, the pruned teacher shows better performance.

Self distillation: Motivated by [11, 55], we conduct the self distillation experiment, where the teacher and the student share the same model. We consider VGG19 and VGG19DBL with CIFAR100 dataset, where ResNet18 and ResNet18DBL are trained on the TinyImageNet dataset. Table 2 shows the test accuracies of 1) the model without KD, 2) the model learned from the unpruned teacher, and 3) the model learned from the pruned teacher. Similar to other experiments, we also observe the consistent result where the pruned model teaches better than the unpruned teacher. Note that learning from unpruned network also increases the test accuracy (compared to the one without a teacher); however, the gain with the pruned teacher is more significant.

Distill ResNet18 to VGG and MobileNet: We also investigate the KD from the pruned teacher when the student and the teacher have different network architectures. Specifically, we consider the TinyImageNet dataset, where the teacher is ResNet18 and students are VGG16 and MobileNetV2 [42]. Table 3 compares the test accuracies of 1) student without a teacher, 2) student learned from the unpruned teacher, and 3) student learned from the

pruned teacher. Consistently, we observe the better KD performance when the teacher is pruned. This implies that the better distillation is not limited to the case of the similar architecture between teacher and student networks. Remark: One might suspect that better distillation result is due to higher accuracy of the teacher, where the pruned model often achieves better accuracy [9]. However, the higher accuracy of the teacher network does not guarantee better results in distillation [46]. Also, the pruned teacher works better even when test accuracy is lower than the unpruned teacher. For example, pruning decreases the test accuracy of the teacher network in ResNet18-TinyImageNet, where we observe that the pruned teacher transfers the knowledge better. This implies that the pruned teacher is better not because it has higher accuracy, but it provides better transferable knowledge. We also investigate the agreement between the teacher and the student's prediction (details provided in Appendix). As shown by Stanton et al. [46], we observe that the agreement and the accuracy behave independently. For example, in VGG19 self distillation experiments, the pruned teacher provides a higher agreement, and the corresponding student has a higher accuracy; however, in ResNet18 self distillation, the pruned teacher shows lower agreement although the student's accuracy is higher. It implies that some students mimic the teacher better but perform worse. This result supports our theory that distillation indirectly helps the training student models with additional regularization.

3.2 Pruned Teacher as a Regularizer

In this section, we provide a theoretical analysis on the pruned teacher in KD. We first point out that the teacher trained with a regularizer provides an additional regularization during distillation. Let $\{(x_i, y_i)\}_{i=1}^N$ be the dataset where the label y_i takes value from the set $\{1, 2, \dots, K\}$. We are interested in a classification model which outputs a K -dimensional probability distributions. Let $f_{\text{true}}(x_i) \in \mathbb{R}^K$ be the one-hot encoded vector where $f_{\text{true}}(x_i)[y_i] = 1$ for the ground-truth label y_i and $f_{\text{true}}(x_i)[y'] = 0$ for all $y' \neq y_i$. We further let $f_t(x; w)$ be the output of the teacher network when the input is x and the weight is w . Then, we train the teacher $f_t(\cdot; w)$ and achieve w_t that minimizes the cross entropy loss $\text{LCE}(w) = \frac{1}{N} \sum_{i=1}^N H(f_{\text{true}}(x_i), f_t(x_i; w))$, (1) where the cross-entropy loss is defined by $H(p_1, p_2) = - \sum_{k=1}^K p_1[k] \log p_2[k]$. Similarly, $f_s(x; \tilde{w})$ is the output of the student network when the input is x and the weight is $\tilde{w}$. For the temperature $\tau = 1$, the knowledge distillation loss is given by $\text{LKD}(\tilde{w}) = \frac{1}{N} \sum_{i=1}^N ((1-\alpha)H(f_{\text{true}}(x_i), f_s(x_i; \tilde{w})) + \alpha H(f_t(x; w_t), f_s(x; \tilde{w})))$. (2) Yuan et al. [55] showed that the KD is equivalent to label smoothing regularization (LSR). More

precisely, the author showed that $LKD(\tilde{w}) = \frac{1}{N} \sum_{i=1}^N H(f(\alpha) m(x_i; w_t), f_s(x_i; \tilde{w}))$, (3) where $f(\alpha) m(x; w_t) = (1-\alpha)f_{true}(x) + \alpha f_t(x; w_t)$, and therefore KD is equivalent to label smoothing regularization with smoothed label distribution $f(\alpha) m(x; w_t)$. We then consider the case where the teacher is trained with a regularizer $R(w)$. The regularized teacher $f_t(\cdot; w_p)$ is obtained by minimizing $LREG(w) = \frac{1}{N} \sum_{i=1}^N H(f_{true}(x_i), f_t(x_i; w)) + R(w)$, (4) i.e., $LREG(w_p) = \min_w LREG(w)$. Since $LCE(w_t) = \min_w LCE(w)$, we have $\frac{1}{N} \sum_{i=1}^N \frac{1}{1 + \exp(-H(f_{true}(x_i), f_t(x_i; w_p)) + R(w_p))} \leq \frac{1}{N} \sum_{i=1}^N \frac{1}{1 + \exp(-H(f_{true}(x_i), f_t(x_i; w_t)) + R(w_t))} \leq \frac{1}{N} \sum_{i=1}^N \frac{1}{1 + \exp(-H(f_{true}(x_i), f_t(x_i; w_t))} \leq \frac{1}{N} \sum_{i=1}^N \log \frac{f_t(x_i; w_t)[y_i]}{f_t(x_i; w_p)[y_i]} \leq R(w_t) - R(w_p)$ (7) Thus, $f_t(x_i; w_t)[y_i]$ is larger than $f_t(x_i; w_p)[y_i]$ on average.

我们通过实验来验证知识挖掘 (KD) 中剪枝教师模型的有效性。我们不是直接对教师网络进行提取（图 1 中的蓝色虚线块），而是先对教师网络进行（非结构化）剪枝，然后再提取到学生网络（图 1 中的红色虚线块）。设置：我们主要考虑使用 VGG [43] 和 ResNet 16 作为教师网络，其中 VGG 在 CIFAR100 数据集 [24] 上训练，ResNet 在 TinyImageNet 数据集 [25] 上训练。TinyImageNet 数据集是经过调整大小的 ImageNet 数据集 6 的一个子集（ $3 \times 64 \times 64$ ）。我们在所有训练中保留 10% 的数据作为验证集。我们应用非结构化剪枝（移除更多权重，更准确地说是 LR 倒带 [39]）来剪枝教师模型。在 LR 回退中，我们将 VGG-CIFAR100 的 epoch 比例设置为 0.65。换句话说，我们先训练 VGG19 200 个 epoch，然后回退学习率并重新训练（微调）网络 130 个 epoch（65%）。需要注意的是，从 0.6 到 0.9 的不同比例在剪枝过程中不会产生显著差异，我们对 ResNet-TinyImageNet 使用 0.5 的比例。为了公平比较，我们训练（和提取）了足够数量的 epoch 的网络，并在验证数据集上达到最佳性能时停止训练。所有测试准确率均为三次独立实验的平均值，并提供了标准差。为简单起见，我们使用原始 KD 20 并设置适当的平衡参数 α 和温度 τ 。平衡参数 α 表示两个目标函数（提取损失和硬目标损失）的比率。温度 τ 是一个软化参数， τ 越高，目标函数越软。在实验中，我们将参数固定为 $\alpha = 0.95$ 和 $\tau = 10$ 。附录中提供了更详细的训练参数。需要注意的是，实验的目的并非达到最佳测试准确率，而是比较从剪枝网络提取和从未剪枝网络提取的结果。因此，超参数以及网络架构并非针对测试准确率进行优化。相反，我们使用原有的设置，以便在剪枝后的教师网络和未剪枝后的教师网络之间进行公平的比较。例如，我们采用针对 ImageNet 数据集 6 优化的 MobileNetV2 [42] 和 ResNet18 16 的默认设置，而我们使用 TinyImageNet 数

数据集 [25]。将 VGG19 提炼为 VGG11：我们将 VGG19 [43] 设置为教师网络，VGG11 设置为学生网络。VGG 的网络架构除了全连接 (FC) 层的数量外，其余均保持不变。我们的 VGG 只有一个 FC 层（通常用于 CIFAR10 数据）。然后，我们比较了常规 VGG19 教师网络和剪枝后的 VGG19 教师网络在 CIFAR100 数据集 [24] 上的知识挖掘 (KD) 结果。我们使用三个稀疏度级别对教师网络进行剪枝：36% 稀疏度（移除 36% 的权重）、59% 稀疏度和 79% 稀疏度。令人惊讶的是，如表 1 所示，使用剪枝后的 VGG19 的 VGG11 的性能始终优于使用未剪枝教师网络的 VGG11。表 1 还提供了当教师网络为 VGG19DBL（每层具有 $2 \times \text{manual}$ 个通道）时的结果。在两种情况下，经过剪枝的教师网络都表现出更好的性能。自我蒸馏：受 [11, 55] 的启发，我们进行了自我蒸馏实验，其中教师网络和学生网络共享同一模型。我们考虑使用 CIFAR100 数据集的 VGG19 和 VGG19DBL，其中 ResNet18 和 ResNet18DBL 在 TinyImageNet 数据集上训练。表 2 展示了 1) 未使用知识挖掘 (KD) 的模型、2) 从未剪枝的教师网络学习的模型以及 3) 从剪枝的教师网络学习的模型的测试准确率。与其他实验类似，我们也观察到一致的结果，即剪枝后的模型比未剪枝的教师网络表现更好。需要注意的是，从未剪枝的网络学习也提高了测试准确率（与没有教师网络的模型相比）；但是，使用剪枝的教师网络的提升更为显著。将 ResNet18 蒸馏到 VGG 和 MobileNet：我们还研究了当学生网络和老师网络采用不同网络架构时，经过剪枝的老师网络的知识交付 (KD)。具体来说，我们考虑了 TinyImageNet 数据集，其中老师网络为 ResNet18，学生网络为 VGG16 和 MobileNetV2 [42]。表 3 比较了以下三种情况的测试准确率：1) 没有老师网络的学生；2) 从未经剪枝的老师网络学习的学生；3) 从剪枝后的老师网络学习的学生。我们一致观察到，老师网络经过剪枝后，知识交付 (KD) 性能更佳。这意味着更好的蒸馏效果并不局限于老师网络和学生网络架构相似的情况。备注：有人可能会认为更好的蒸馏结果是由于老师网络的准确率更高，而剪枝后的模型通常能获得更好的准确率 [9]。然而，老师网络的准确率更高并不能保证蒸馏结果更好 [46]。此外，即使测试准确率低于未剪枝的教师网络，剪枝后的教师网络也能更好地工作。例如，在 ResNet18-TinyImageNet 中，剪枝会降低教师网络的测试准确率，而

我们观察到，经过剪枝的教师模型能够更好地迁移知识。这意味着，经过剪枝的教师模型之所以更好，并不是因为它拥有更高的准确率，而是因为它提供了更好的可迁移知识。我们还研究了教师模型与学生模型预测之间的一致性（详情见附录）。正如 Stanton 等人 [46] 的研究所示，我们观察到一致性和准确率之间相互独立。例如，在 VGG19 自蒸馏实验中，经过剪枝的教师模

型提供了更高的一致性，相应的学生模型也具有更高的准确率；然而，在 ResNet18 自蒸馏实验中，尽管学生模型的准确率更高，但经过剪枝的教师模型的一致性却较低。这意味着有些学生模型模仿得更好，但实际表现更差。这一结果支持了我们的理论，即蒸馏通过额外的正则化间接地帮助学生模型的训练。

3.2 经过剪枝的教师模型作为正则化器

在本节中，我们将对知识蒸馏中的经过剪枝的教师模型进行理论分析。我们首先指出，使用正则化器训练的教师模型在蒸馏过程中提供了额外的正则化。设 $\{(x_i, y_i)\}_{i=1}^N$ 为数据集，其中标签 y_i 从集合 $\{1, 2, \dots, K\}$ 中取值。我们感兴趣的是一个输出 K 维概率分布的分类模型。设 $f_{\text{true}}(x_i) \in \mathbb{R}^K$ 为独热编码向量，其中 $f_{\text{true}}(x_i)[y_i] = 1$ 为真实标签 y_i ， $f_{\text{true}}(x_i)[y'] = 0$

对于所有 $y' \neq y_i$ 。我们进一步令 $f_t(x; w)$ 为教师网络的输出，当输入为 x ，权重为 w 时。然后，我们训练教师网络 $f_t(\cdot; w)$ ，并得到使交叉熵损失最小化的 w_t ，类似地， $f_s(x; \tilde{w})$ 为学生网络的输出，当输入为 x ，权重为 $\tilde{w}$ 时。对于温度 $\tau = 1$ ，知识蒸馏损失 Yuan 等人 [55] 证明知识蒸馏 (KD) 等价于标签平滑正则化 (LSR)。更准确地说，作者证明了其中 $f(\alpha) m(x; w_t) = (1-\alpha)f_{\text{true}}(x) + \alpha f_t(x; w_t)$ ，因此知识蒸馏 (KD) 等价于具有平滑标签分布 $f(\alpha) m(x; w_t)$ 的标签平滑正则化。然后，我们考虑使用正则化器 $R(w)$ 训练教师模型的情况。正则化教师模型 $f_t(\cdot; w_p)$ 可通过最小化因此， $f_t(x_i; w_t)[y_i]$ 平均大于 $f_t(x_i; w_p)[y_i]$ 。

We study the problem of distilling knowledge from a large deep teacher network to a much smaller student network for the task of road marking segmentation. In this work, we explore a novel knowledge distillation (KD) approach that can transfer ‘knowledge’ on scene structure more effectively from a teacher to a student model. Our method is known as Inter-Region Affinity KD (IntRA-KD). It decomposes a given road scene image into different regions and represents each region as a node in a graph. An inter-region affinity graph is then formed by establishing pairwise relationships between nodes based on their similarity in feature distribution. To learn structural knowledge from the teacher network, the student is required to match the graph generated by the teacher. The proposed method shows promising results on three large-scale road marking segmentation benchmarks, i.e., ApolloScape, CU-Lane and LLAMAS, by taking various lightweight models as students and ResNet-101 as the teacher. IntRA-KD consistently brings higher performance gains on all lightweight models, compared to previous distillation methods.

Road marking segmentation. Road marking segmentation is conventionally handled using hand-crafted features to obtain road marking segments. Then, a classification network is employed to classify the category of each segment [10, 19]. These approaches have many drawbacks, e.g., require sophisticated feature engineering process and only work well in simple highway scenarios. The emergence of deep learning has avoided manual feature design through learning features in an end-to-end manner. These approaches usually adopt the dense prediction formulation, i.e., assign each pixel a category label [8, 14, 24]. For example, Wang et al. [24] exploit deep neural networks to map an input image to a segmentation map. Since large models usually demand huge memory storage and have slow inference speed, many lightweight models, e.g., ERFNet [20], are leveraged to fulfil the requirement of fast inference and small storage [8]. However, due to the limited model size, these small networks perform poorly in road marking segmentation. A common observation is that such small models do not have enough capacity to capture sufficient contextual knowledge given the sparse supervision signals [8, 14, 29]. Several schemes have been proposed to relieve the sparsity problem. For instance, Hou et al. [8] reinforce the learning of contextual knowledge through self knowledge distillation, i.e., using deep-layer attention maps to guide the learning of shallower layers. SCNN [14] resolves this problem through message passing between deep feature layers. Zhang et al. [29] propose a framework to perform lane area segmentation and lane boundary detection simultaneously. The aforementioned methods do not take structural relationship between different areas into account and they do not consider knowledge distillation from teacher networks.

Knowledge distillation. Knowledge distillation was originally introduced by [6] to transfer knowledge from a teacher model to a compact student model. The distilled knowledge can be in diverse forms, e.g., softened output logits [6], intermediate feature maps [4, 9, 13, 31] or pairwise similarity maps between neighbouring layers [26]. There is another line of work [8, 22] that uses self-derived knowledge to reinforce the representation learning of the network itself, without the supervision of a large teacher model. Recent studies have expanded knowledge distillation from one sample to several samples [12, 15, 17, 23]. For instance, Park et al. [15] transfer mutual relations between a batch of data samples in the distillation process. Tung et al. [23] take the similarity scores of features of different samples as distillation targets. The aforementioned approaches [12, 15, 17, 23] do not consider the structural relationship between different areas in one

sample. On the contrary, the proposed IntRA-KD takes inter-region relationship into account, which is new in knowledge distillation.

Tables 2-4 summarize the performance of our method, i.e., ERFNet-IntRA-KD, against state-of-the-art algorithms on the testing set of ApolloScape 11, CULane 14 and LLAMAS 1. We also report the runtime and parameter size of different models in Table 3. The runtime is recorded using a single GPU (GeForce GTX TITAN X Maxwell) and averages across 100 samples. ERFNet-IntRA-KD outperforms all baselines and previous distillation methods in all three benchmarks. Note that ERFNet-IntRA-KD has $21 \times$ fewer parameters and runs $16 \times$ faster compared with ResNet-101 on CULane testing set; the appealing performance strongly suggests the effectiveness of IntRA-KD. We also apply IntRA-KD to ENet and ResNet-18, and find that IntRA-KD can equivalently bring more performance gains to the backbone models than the state-of-the-art BiFPN [31] on ApolloScape dataset (Fig. 7). Note that BiFPN is a competitive algorithm in all benchmarks. Encouraging results are also observed on CULane and LLAMAS when applying IntRA-KD to ENet and ResNet-18. Due to space limit, we report detailed performance of applying different distillation algorithms to ENet and ResNet-18 in the supplementary material. The effectiveness of our IntRA-KD on different backbone models validates the good generalization ability of our method. We also show some qualitative results of our IntRA-KD and BiFPN [31] (the most competitive baseline) on three benchmarks. As shown in (a) and (c) of Fig. 8, IntRA-KD helps ERFNet predict more accurately on both long and thin road markings. As to other challenging scenarios of crowded roads and poor light conditions, ERFNet and ERFNet-BiFPN either predict lanes inaccurately or miss the predictions. By contrast, predictions yielded by ERFNet-IntRA-KD are more complete and accurate. Apart from model predictions, we also show the deep feature embeddings of different methods. As can be seen from Fig. 9, the embedding of ERFNet-IntRA-KD is more structured compared with that of ERFNet and ERFNet-BiFPN. In particular, the features of ERFNet-IntRA-KD are more distinctly clustered according to their classes in the embedding, with similar distribution to the embedding of the ResNet-101 teacher. The results suggest the importance of structural information in knowledge distillation.

We have proposed a simple yet effective distillation approach, i.e., IntRA-KD, to effectively transfer scene structural knowledge from a teacher model to a student

model. The structural knowledge is represented as an inter-region affinity graph to capture similarity of feature distribution of different scene regions. We applied IntRA-KD to various lightweight models and observed consistent performance gains to these models over other contemporary distillation methods. Promising results on three large-scale road marking segmentation benchmarks strongly suggest the effectiveness of IntRA-KD. Results on Cityscapes are provided in the supplementary material.

我们研究了将知识从大型深度教师网络提炼到规模小得多的学生网络以完成道路标记分割任务的问题。在本研究中，我们探索了一种新颖的知识提炼 (KD) 方法，该方法可以更有效地将场景结构的“知识”从教师模型迁移到学生模型。我们的方法被称为区域间亲和性知识提炼 (IntRA-KD)。它将给定的道路场景图像分解为不同的区域，并将每个区域表示为图中的节点。然后，通过基于节点在特征分布上的相似性建立节点间的成对关系，构建区域间亲和性图。为了从教师网络中学习结构性知识，学生需要匹配教师生成的图。该方法以各种轻量级模型为学生，ResNet-101 为老师，在三个大规模道路标记分割基准数据集（即 ApolloScape、CU-Lane 和 LLAMAS）上取得了良好的效果。与之前的蒸馏方法相比，IntRA-KD 在所有轻量级模型上均能持续带来更高的性能提升。

道路标记分割。道路标记分割传统上采用手工设计特征来获取道路标记段。然后，使用分类网络对每个段进行分类 [10, 19]。这些方法存在许多缺点，例如，需要复杂的特征工程流程，并且仅适用于简单的高速公路场景。深度学习的出现通过端到端的特征学习避免了手工特征设计。这些方法通常采用密集的预测公式，即为每个像素分配一个类别标签 [8, 14, 24]。例如，Wang 等人 [24] 利用深度神经网络将输入图像映射到分割图。由于大型模型通常需要巨大的内存存储空间并且推理速度较慢，因此许多轻量级模型（例如 ERFNet 20）被用来满足快速推理和小存储的需求 8。然而，由于模型规模有限，这些小型网络在道路标记分割方面表现不佳。一个常见的观察结果是，在稀疏的监督信号下，此类小型模型没有足够的能力来捕获足够的上下文知识[8, 14, 29]。为了缓解稀疏性问题，已经提出了几种方案。例如，Hou 等人 8 通过自知识蒸馏来强化上下文知识的学习，即使用深层注意力图来指导较浅层的学习。SCNN 14 通过深层特征层之间的消息传递解决了这个问题。Zhang 等人[29]提出了一个同时进行车道区域分割和车道边界检测的框架。上述方法没有考虑不同区域之间的结构关系，也没有考虑从教师网络中进行知识蒸馏。知识蒸馏。

知识蒸馏最初由 [6] 提出，用于将知识从教师模型迁移到紧凑的学生模型。蒸馏后的知识可以有多种形式，例如软化的输出逻辑值 [6]、中间特征图 [4, 9, 13, 31] 或相邻层之间的成对相似性图 [26]。另一项研究 [8, 22] 使用自衍生知识来强化网络自身的表征学习，而无需大型教师模型的监督。最近的研究将知识蒸馏的范围从单个样本扩展到多个样本 [12, 15, 17, 23]。例如，Park 等人 [15] 在蒸馏过程中迁移一批数据样本之间的相互关系。Tung 等人 [23] 将不同样本特征的相似度得分作为蒸馏目标。上述方法 [12, 15, 17, 23] 没有考虑样本内不同区域之间的结构关系，而我们提出的 IntRA-KD 方法考虑了区域间的关系，这在知识提炼方面是一个创新。

表 2-4 总结了我们的方法 ERFNet-IntRA-KD 在 ApolloScape [11]、CULane [14] 和 LLAMAS [1] 测试集上与当前最佳算法的性能比较。我们还在表 3 中报告了不同模型的运行时间和参数大小。运行时间使用单块 GPU (GeForce GTX TITAN X Maxwell) 记录，并对 100 个样本取平均值。ERFNet-IntRA-KD 在所有三个基准测试中均优于所有基线和之前的蒸馏方法。值得注意的是，在 CULane 测试集上，与 ResNet-101 相比，ERFNet-IntRA-KD 的参数减少了 21 倍，运行速度提高了 16 倍；如此优异的性能有力地证明了 IntRA-KD 的有效性。我们还将 IntRA-KD 应用于 ENet 和 ResNet-18，发现在 ApolloScape 数据集上，IntRA-KD 可以为骨干模型带来比最先进的 BiFPN [31] 更高的性能提升（图 7）。需要注意的是，BiFPN 在所有基准测试中都表现出色。将 IntRA-KD 应用于 ENet 和 ResNet-18 时，在 CULane 和 LLAMAS 上也观察到了令人鼓舞的结果。由于篇幅限制，我们在补充材料中详细报告了将不同的蒸馏算法应用于 ENet 和 ResNet-18 的性能。IntRA-KD 在不同骨干模型上的有效性证明了我们方法的良好泛化能力。我们还展示了 IntRA-KD 和 BiFPN [31]（最具竞争力的基准）在三个基准测试上的一些定性结果。如图 8 (a) 和 (c) 所示，IntRA-KD 可帮助 ERFNet 更准确地预测长而细的道路标记。对于拥挤的道路和光线不足等其他具有挑战性的场景，ERFNet 和 ERFNet-BiFPN 要么对车道预测不准确，要么预测失败。相比之下，ERFNet-IntRA-KD 的预测更加完整和准确。除了模型预测之外，我们还展示了不同方法的深度特征嵌入。从图 9 可以看出，与 ERFNet 和 ERFNet-BiFPN 相比，ERFNet-IntRA-KD 的嵌入更具结构性。特别是，ERFNet-IntRA-KD 的特征在嵌入中根据其类别更明显地聚类，其分布与 ResNet-101 教师网络的嵌入相似。结果表明了结构信息在知识提炼中的重要性。

我们提出了一种简单有效的知识蒸馏方法，即 IntRA-KD，用于有效地将场景结构知识从教师模型迁移到学生模型。结构知识以区域间亲和图的形式表示，以捕捉不同场景区域特征分布的相似性。我们将 IntRA-KD 应用于各种轻量级模型，并观察到这些模型的性能相对于其他当代知识蒸馏方法而言始终如一地提升。在三个大规模道路标记分割基准测试中取得的良好结果有力地证明了 IntRA-KD 的有效性。Cityscapes 数据集上的结果已在补充材料中提供。

Knowledge distillation aims at transferring knowledge acquired in one model (a teacher) to another model (a student) that is typically smaller. Previous approaches can be expressed as a form of training the student to mimic output activations of individual data examples represented by the teacher. We introduce a novel approach, dubbed relational knowledge distillation (RKD), that transfers mutual relations of data examples instead. For concrete realizations of RKD, we propose distance-wise and angle-wise distillation losses that penalize structural differences in relations. Experiments conducted on different tasks show that the proposed method improves educated student models with a significant margin. In particular for metric learning, it allows students to outperform their teachers' performance, achieving the state of the arts on standard benchmark datasets.

In this section we first revisit conventional KD and introduce a general form of RKD. Then, two simple yet effective distillation losses will be proposed as instances of RKD. RKD aims at transferring structural knowledge using mutual relations of data examples in the teacher's output presentation. Unlike conventional approaches, it computes a relational potential ψ for each n -tuple of data examples and transfers information through the potential from the teacher to the student.

We evaluate RKD on three different tasks: metric learning, classification, and few-shot learning. Throughout this section, we refer to RKD with the distance-wise loss as RKD-D, that with angle-wise loss as RKD-A, and that with two losses together as RKD-DA. When the proposed losses are combined with other losses during training, we assign respective balancing factors to the loss terms. We compare RKD with other KD methods, e.g., FitNet [27], Attention [47] and HKD (Hinton's KD) [11]. For metric learning, we conduct an additional comparison with DarkRank [7] which is a KD method specifically designed for metric learning. For fair comparisons, we tune hyperparameters of the competing methods using grid search. We have demonstrated on different tasks and benchmarks that the proposed RKD effectively

transfers knowledge using mutual relations of data examples. In particular for metric learning, RKD enables smaller students to even outperform their larger teachers.

While the distance-wise and angle-wise distillation losses used in this work turn out to be simple yet effective, the RKD framework allows us to explore a variety of task-specific RKD losses with high-order potentials beyond the two instances. We believe that the RKD framework opens a door to a promising area of effective knowledge transfer with high-order relations.

知识蒸馏旨在将一个模型（教师模型）获得的知识迁移到另一个通常规模较小的模型（学生模型）。以往的方法可以表示为训练学生模型模仿教师模型所代表的单个数据示例的输出激活。我们提出了一种名为关系知识蒸馏

（RKD）的新方法，它迁移的是数据示例之间的相互关系。对于 RKD 的具体实现，我们提出了基于距离和角度的蒸馏损失函数，用于惩罚关系的结构性差异。在不同任务上进行的实验表明，该方法能够显著提升已训练的学生模型。尤其是在度量学习方面，它能够使学生的表现超越教师模型，在标准基准数据集上达到最佳水平。

在本节中，我们首先回顾传统的知识提升 (KD)，并介绍一种通用的学习损失 (RKD)。然后，我们将提出两个简单但有效的蒸馏损失作为 RKD 的实例。RKD 旨在利用教师输出呈现中数据示例的相互关系来迁移结构性知识。与传统方法不同，它为每个 n 元组数据示例计算一个关系势 ψ ，并通过该势将信息从教师传递给学生。

我们在三个不同的任务上评估 RKD：度量学习、分类和小样本学习。在本节中，我们将包含距离损失的 RKD 称为 RKD-D，包含角度损失的 RKD 称为 RKD-A，包含两个损失的 RKD 称为 RKD-DA。当在训练过程中将提出的损失与其他损失组合时，我们会为损失项分配相应的平衡因子。我们将 RKD 与其他知识挖掘 (KD) 方法进行了比较，例如 FitNet [27]、Attention [47] 和 HKD (Hinton 的知识挖掘) [11]。对于度量学习，我们与专为度量学习而设计的知识挖掘 (KD) 方法 DarkRank [7] 进行了额外的比较。为了公平起见，我们使用网格搜索调整了竞争方法的超参数。我们已经在不同的任务和基准测试中证明了，所提出的 RKD 能够有效地利用数据示例的相互关系进行知识迁移。特别是在度量学习方面，RKD 能够使体型较小的学生甚至超越体型较大的老师。虽然本研究使用的距离和角度蒸馏损失函数被证明简单而有效，但 RKD 框架使我们能够探索除上述两种情况之外，各种特定于任务的、

具有高阶潜力的 RKD 损失函数。我们相信，RKD 框架为利用高阶关系进行有效知识转移打开了一扇充满希望的大门。

The ability to perform pixel-wise semantic segmentation in real-time is of paramount importance in mobile applications. Recent deep neural networks aimed at this task have the disadvantage of requiring a large number of floating point operations and have long run-times that hinder their usability. In this paper, we propose a novel deep neural network architecture named ENet (efficient neural network), created specifically for tasks requiring low latency operation. ENet is up to $18\times$ faster, requires $75\times$ less FLOPs, has $79\times$ less parameters, and provides similar or better accuracy to existing models. We have tested it on CamVid, Cityscapes and SUN datasets and report on comparisons with existing state-of-the-art methods, and the trade-offs between accuracy and processing time of a network. We present performance measurements of the proposed architecture on embedded systems and suggest possible software improvements that could make ENet even faster.

在移动应用中，实时执行像素级语义分割的能力至关重要。目前用于此任务的深度神经网络存在需要大量浮点运算且运行时间过长的缺点，这限制了它们的可用性。本文提出了一种名为 ENet（高效神经网络）的新型深度神经网络架构，专为需要低延迟运行的任务而设计。ENet 的速度提高了 18 倍，所需的浮点运算次数减少了 75 倍，参数量减少了 79 倍，并且准确率与现有模型相当甚至更高。我们已经在 CamVid、Cityscapes 和 SUN 数据集上对其进行了测试，并报告了与现有最先进方法的比较结果，以及网络准确率和处理时间之间的权衡。我们展示了所提出的架构在嵌入式系统上的性能测量结果，并提出了一些可能使 ENet 更快的软件改进建议。

The obstacle detection field is a very broad one and a lot of obstacle detection systems have been developed in the last years in this domain. We tried to identify the main character of an obstacle detection system from the cluttered scene. Thus, we classified the main types of sensors from this field in passive (visible and infrared spectrum camera) and active (radar, laser-scanner, sonar) sensors and we made a survey in this domain. After a short presentation of every type of sensor, we presented another current and fancy solution for an obstacle detection system: the fusion of different sensors together. Almost all obstacle detection systems use a combination of passive-active technology, and in general the best solution is obtained using a vision system combined with a distance sensor like radar or laser. Maybe the most low-priced system

is one combining only vision systems, but the inconvenient in this case is the lack of distance information.

Vision systems Video sensors provide a high spatial resolution, both horizontally and vertically. Almost all approaches using vision implies two steps: object detection followed by object recognition. In the detection step a region of interest (ROI) (e.g. a rectangular bounding box) is found and it is associated with a potential obstacle; then, the recognition process follows, where the type of objects is determined. Most often an object detection system is formed from three main components: one module which produces the segmentation of the objects from the background, another one which classify the objects and the last module which track the objects. Segmentation is a method to extract objects or features of the objects from the background. There are three segmentation methods used in the obstacle detection domain: stereo, motion or infrared based segmentation. The stereoscopy is a new developed technique used to extract three-dimensional information from bi- dimensional images. The extraction of depth information (also called stereo matching) is a process of finding correspondences between two correlated images 1, 2. The main limit of the motion based segmentation technique is the need of processing a number of successive frames which increases the processing time 3. The information extracted after segmentation can be a model or some features of the object. The system must be learned a priori with some features in order to detect and recognize these features. From the obtained images certain features are extracted (which will represent the pattern), and these patterns will be essentials for the learning and testing process. An autonomous vehicle combines information provided from multiple sensors in order to obtain a better cognition of the surrounded. In this case, the provided vision data should come from: different sensors (visible and infrared), or from the same sensor (video or stereo images but taken at different moment of time or space) and they should be correlated each other. Different cameras or configurations are used in vision-based obstacle detection systems: One possibility is that camera be in move (installed on a moving vehicle) or static (for surveillance purposes 4). For video frames captured in intelligent vehicles the frame-background are always changing and unpredictable, and therefore no reference background (as in the case of static camera) can be used due to the motion of the objects inside the scene and the movement of the camera. Systems using moving cameras in visible, and in infrared domain are presented in [5,6,7,8] and [9,10,11]. The second possibility is the use of a single

image or a sequence of images (video images). Single images are very rarely used alone for obstacle detection applications. The time processing is not very expensive, but either the performances of such a system are not very good. To recognize obstacles in single images can be a problem, because such a system is not using either the motion or stereo information. This is the case in which any obstacle from image is seen as a still object and for his detection shape-based approaches are used. Systems in which objects symmetry properties are exploited can be found in visible [5,12] and infrared [10,13] domain. Video images (or sequences of consecutive frames at increased moment of time) are processed in order to provide a motion vector, needed to identify moving objects. A third possibility is the use of monocular or stereo images. Monocular images are often used for shape-based approaches in the visible domain [5,12,14] and in infrared domain [9,10,13]. This kind of images is also used in motion-based approaches in visible 15 and infrared 16 domain. One alternative solution to the monocular images is the use of stereo images, in visible [6,17] and infrared 11 domain. Different algorithms are proposed to detect pedestrians in images acquired from far infrared (FIR) cameras [9,10,13,16]. Honda has developed an intelligent night vision system using two far infra-red cameras (one stereo system) which were installed in the front bumper of the vehicle. Vision-based technique is still the easiest and utilized method to analyze the road, but more and more works appear in this domain and several of sensors-combination systems were developed. Almost all obstacle-detection systems use visible spectrum or infrared cameras in fusion with an active sensor, like radar, laser scanner or sonar to increase the performances of such a system. Systems using active sensors The use of radar for obstacle detection has advantages especially in bad weather situations. Radar sensor measures only point targets of objects like cars, trees, traffic signs, bicycles and human beings while other objects which does not have a high reflectance, like street and pavement, are not detected. In 18 and 19 some radar parameters like RCS (Radar Cross Section), power spectral density (PSD) and object's velocity coming from radar are registered (for pedestrians and vehicles) and compared in order to differentiate between these two types of objects.

The laser scanner measures the outline of objects with high resolution and accuracy. Object velocity is derived from the positional information of each scan 19. The original data from a laser scanner is much like vision image. Clustered data points are grouped into different objects using segmentation and then the objects are classified

into different categories (pedestrians, cars, trucks, bicycles) according to their characteristics. Some laser scanner characteristics are presented in 20, 21.

Sensorfusion A much exploited idea is to combine some different sensors inputs together. In that way, one technology's advantage can be used to compensate another's limitation; the combination process is often called "sensor fusion". This sensorial fusion is done by humans every day, since humans combine visual with acoustic and tactile information to get more knowledge about the surrounding. Many times, the use of one single sensor is not very useful for an obstacle-detection system, because there is no such a perfect sensor which can provide rich information about the surrounding in any weather conditions and in any illumination situation. So it is a natural idea to make use of the advantages of one sensor to suppress another's disadvantages. A fusion process can take place on different levels: low or high level. Low level (data-level) means that the fusion is made at data information and combines results before a hard decision is made. High level fusion means that the fusion is established on a list of objects. It combines results of different modules after some decisions are made about the size and position of the obstacle. In general, in a high-level fusion, a list of objects is generated with one sensor and it is checked with help of the other sensor. Using different sensor technologies together, the system performances can be significantly improved. The sensor fusion increases the evaluation speed by reducing the search space (region of interest). Thus, one or more sensors of different technologies are bring together to perform a better recognition of the object. In general, because active sensors offer limited spatial information in opposite with a vision system which has rich information regarding visual appearance of an object, a fusion between passive and active sensors is desired. Radar offer good accuracy in longitudinal distance measurement, but poor accuracy in lateral distance measurement and because vision has opposite properties, a fusion will bring good accuracy in both positional measurements. In 21 and 22 radar and vision sensors are used for a fusion application in dense city traffic situations. In 23 the sensor system for vehicles detection consists of a MMW (Milli- Meter Wave) radar and a camera sensor fusioned using a Bayesian network. Target car image is clipped out from whole image by using object position data acquired by MMW radar to reduce computational load of image processing. The main advantage of using IR camera instead of visible spectrum camera is that the segmentation problem is simplified. Sensors fusion using IR cameras and radar is suggested to support the driver in reduced visibility, due to

night and adverse weather conditions in [24]. For the obstacle detection system presented in [25] data fusion is performed between stereovision and laser scanner. In 20 a multi sensor system that uses an infrared camera with a laser scanner is used. To handle the information from different sensors a Kalman filter based data fusion is used. Where a single radar sensor installed on a vehicle may not be enough to cover the whole interested area, a radar network can be used. A high performance sensor platform for pedestrians and cyclists protection is presented in [26]. Their data come from: a radar network composed of several 12GHz radar sensors, an image system composed of passive infrared (IR) and an visible color vision systems is fused in a low and high level data fusion technique. A pedestrian detection system from a moving camera is proposed in 18. The fusion of a radar system consisting of two individual 12 GHz radar sensors and a monocular vision system was performed. A target-list is generated from radar and the remained filtered pedestrian candidates are used as inputs for the signal- processing module of the vision sensor. VI.

CONCLUSION A complete and reliable sensing system for obstacle detection can benefit a lot from the combined use of multiple types of sensors, especially from the active-passive combination. Any one particular type of technology may have difficulties to meet all necessary requirements in order to detect an obstacle in various lighting or weather conditions. The clutter background and complex moving patterns of all objects which may appear on a road scene demand sophisticated processing of sensor inputs.

障碍物检测领域非常广泛，近年来该领域已开发出许多障碍物检测系统。我们试图从更复杂的场景中识别障碍物检测系统的主要特征。因此，我们将该领域的主要传感器类型分为被动（可见光和红外光谱摄像机）和主动（雷达、激光扫描仪、声纳）传感器，并对该领域进行了研究。在简要介绍每种类型的传感器之后，我们提出了另一种当前流行的障碍物检测系统解决方案：不同传感器的融合。几乎所有障碍物检测系统都采用了主动-被动技术的组合，通常，最佳解决方案是将视觉系统与雷达或激光等距离传感器相结合。也许最经济的系统是仅结合视觉系统的系统，但这种情况下存在的不足是缺乏距离信息。

视觉系统 视频传感器在水平和垂直方向上均提供高空间分辨率。几乎所有使用视觉的方法都包含两个步骤：物体检测和物体识别。在检测步骤中，找到一个感兴趣区域 (ROI)（例如，矩形边界框），并将其与潜在障碍物关联；然后，进行识别过程，确定物体的类型。通常，一个物体检测系统由三个主要组

件组成：一个模块用于从背景中分割物体，另一个模块用于对物体进行分类，最后一个模块用于跟踪物体。分割是一种从背景中提取物体或物体特征的方法。障碍物检测领域使用三种分割方法：立体分割、运动分割或基于红外的分割。立体视觉是一种新兴的技术，用于从二维图像中提取三维信息。深度信息的提取（也称为立体匹配）是在两幅相关图像之间寻找对应关系的过程 1, 2。基于运动的分割技术的主要局限性在于需要处理大量连续帧，这会增加处理时间 3。分割后提取的信息可以是物体的模型或某些特征。系统必须先验地学习某些特征才能检测和识别这些特征。从获取的图像中提取某些特征（这些特征将代表模式），这些模式对于学习和测试过程至关重要。自动驾驶汽车会结合来自多个传感器的信息，以便更好地了解周围环境。在这种情况下，提供的视觉数据应该来自：不同的传感器（可见光和红外），或来自同一传感器（在不同时间或空间时刻拍摄的视频或立体图像），并且它们应该相互关联。基于视觉的障碍物检测系统使用不同的摄像机或配置：一种可能性是摄像机处于移动状态（安装在移动车辆上）或静止状态（用于监控目的 4）。对于智能车辆捕获的视频帧，帧背景始终在变化且不可预测，因此由于场景内物体的运动以及摄像机的运动，无法使用参考背景（如静态摄像机的情况）。[5,6,7,8] 和 [9,10,11] 介绍了在可见光和红外域中使用移动摄像机的系统。第二种可能性是使用单幅图像或一系列图像（视频图像）。单幅图像很少单独用于障碍物检测应用。处理时间不是很昂贵，但这种系统的性能不是很好。识别单幅图像中的障碍物可能是一个问题，因为这样的系统没有使用运动或立体信息。在这种情况下，图像中的任何障碍物都被视为静止物体，并使用基于形状的方法进行检测。在可见光[5,12]和红外[10,13]领域可以找到利用物体对称性特性的系统。视频图像（或连续帧序列，在增加的时间点）经过处理，可以提供识别运动物体所需的运动矢量。第三种可能性是使用单目图像或立体图像。单目图像通常用于可见光[5,12,14]和红外[9,10,13]领域基于形状的方法。这类图像也用于可见光 15 和红外 16 领域基于运动的方法。单目图像的另一种解决方案是在可见光 [6,17]和红外 11 领域使用立体图像。已提出了不同的算法来检测远红外 (FIR) 摄像机获取的图像中的行人 [9,10,13,16]。本田开发了一套智能夜视系统，该系统采用安装在车辆前保险杠上的两个远红外摄像机（其中一个为立体系统）。基于视觉的技术仍然是分析道路最简单、最常用的方法，但该领域的研究成果越来越多，并开发了多种传感器组合系统。几乎所有障碍物检测系统都使用可见光谱或红外摄像机与主动传感器（如雷达、激光扫描仪或声纳）融合，以提高此类系统的性能。使用主动传感器的系统使用雷达进行障碍物检测具有优势，尤其是在恶劣天气条件下。雷达传感器仅测量汽车、树木、交通标志、自

行车和人体等点目标物体，而无法检测到其他反射率不高的物体，如街道和人行道。在 18 和 19 中，一些雷达参数，如 RCS（雷达截面）、功率谱密度（PSD）和来自雷达的物体速度，都记录在案（针对行人和车辆）进行分类并进行比较，以区分这两类物体。

激光扫描仪能够以高分辨率和高精度测量物体轮廓。物体速度由每次扫描的位置信息推导出来 19。激光扫描仪的原始数据与视觉图像非常相似。聚类的数据点通过分割技术被分组到不同的物体中，然后根据物体的特征将其分类为不同的类别（行人、汽车、卡车、自行车）。20 和 21 中介绍了一些激光扫描仪的特性。

传感器融合 一个被广泛应用的想法是将一些不同的传感器输入组合在一起。这样，一种技术的优势可以用来弥补另一种技术的局限性；这种组合过程通常被称为“传感器融合”。人类每天都在进行这种传感器融合，因为人类将视觉、听觉和触觉信息结合起来，以获得更多关于周围环境的知识。很多时候，单一传感器对于障碍物检测系统来说效果并不理想，因为没有哪个传感器能够完美地提供丰富的环境信息，无论天气条件还是光照条件。因此，利用一个传感器的优势来抵消另一个传感器的劣势是一个自然的想法。融合过程可以分为低级和高级两种。低级（数据级）意味着融合基于数据信息进行，并在做出硬决策之前合并结果。高级融合意味着融合基于一系列物体。在对障碍物的大小和位置做出一些决策后，它会合并不同模块的结果。通常，在高级融合中，一个传感器会生成一个物体列表，并借助另一个传感器进行检查。结合使用不同的传感器技术，可以显著提升系统性能。传感器融合通过缩小搜索空间（感兴趣区域）来提高评估速度。因此，将一个或多个不同技术的传感器组合在一起，可以更好地识别物体。一般而言，由于有源传感器提供的空间信息有限，而视觉系统则拥有丰富的物体外观信息，因此需要将无源传感器和有源传感器进行融合。雷达在纵向距离测量方面精度较高，但在横向距离测量方面精度较差，而视觉具有相反的特性，因此融合可以在两种位置测量中都带来良好的精度。在 21 和 22 中，雷达和视觉传感器被用于密集城市交通环境中的融合应用。在 23 中，用于车辆检测的传感器系统由毫米波（MMW）雷达和摄像头传感器组成，并使用贝叶斯网络进行融合。利用毫米波雷达获取的物体位置数据，从整幅图像中裁剪出目标汽车图像，以减少图像处理的计算负荷。使用红外摄像头替代可见光谱摄像头的主要优势在于简化了分割问题。[24] 建议使用红外摄像头和雷达进行传感器融合，以便在夜间和恶劣天气条件下，为驾驶员提供能见度较低的车辆辅助。[25] 提出的障碍物检测系统在立体视觉和激光扫描仪之间进行了数据融合。20 提出了一种多传感器系统，该系统结合了红外摄像头和激光扫描仪。为了处理来自不同传感器的信息，采用了基于卡尔曼滤波

器的数据融合方法。当车辆上安装的单个雷达传感器不足以覆盖整个目标区域时，可以使用雷达网络。[26] 提出了一种用于保护行人和骑车人的高性能传感器平台。其数据来源于：由多个 12GHz 雷达传感器组成的雷达网络、由被动红外 (IR) 组成的图像系统以及可见彩色视觉系统，并通过低级和高级数据融合技术进行融合。18 提出了一种基于移动摄像机的行人检测系统。该系统融合了由两个独立的 12 GHz 雷达传感器组成的雷达系统和单目视觉系统。雷达生成目标列表，并将筛选出的行人候选列表作为视觉传感器信号处理模块的输入。VI. 结论 一个完整可靠的障碍物检测传感系统可以从多种类型传感器的组合使用中获益良多，尤其是主被动传感器的组合。任何一种特定类型的技术都可能难以满足在各种光照或天气条件下检测障碍物的所有必要要求。道路场景中可能出现的各种物体的杂乱背景和复杂的运动模式要求对传感器输入进行复杂的处理。

This paper presents a keypoint detection method based on the Laplacian of Gaussian (LoG). In contrast to the Difference of Gaussian (DoG)-based keypoint detection method used in Scale Invariant Feature Transform (SIFT), we focus on the LoG operator and its higher order derivatives. We provide mathematical analogies between higher order DoG (HDoG) and higher order LoG (HLoG) and experimental results to show the effectiveness of the proposed HLoG-based keypoint detection method. The performance of the HLoG is evaluated with four different tests: i) a repeatability test of the keypoints detected across images under various transformations, ii) image retrieval, iii) panorama stitching and iv) 3D reconstruction. The proposed HLoG method provides comparable performance to HDoG and the combination of HLoG and HDoG provides significant improvements in various keypoint-related computer vision problems.

In conclusion, we proposed an alternative keypoint detector based on higher order spatial derivatives, HLoG. We showed the effectiveness of the proposed method based on four different experimental evaluations. Firstly, a repeatability test was performed using a small set of images with known homography information. Secondly, an image retrieval test was performed using a data set with a larger number of images. Thirdly, a panoramic stitching test was performed to show the effectiveness of the proposed method in the image alignment problem. Finally, we performed a 3D reconstruction test and verified that the proposed method helped in generating a larger number of 3D points. We have successfully shown that the

proposed HLoG-based keypoint detection method is advantages when it is used together with HDoG for various computer vision problems. We believe that the improvement in performance with the fusion of HDoG and HLoG is due to the difference between the base operators (i.e., DoG vs. LoG). The differences between HDoG and HLoG add up in each derivation order and become significant when multiple derivation orders are considered. We will further evaluate HDoG and HLoG to understand their characteristics and optimal parameter settings. We are optimizing and trying to apply a normalization technique to the HLoG filters to stabilize the filter response to high frequency components. We will also apply HLoG to more computer vision problems such as 2D or 3D alignments and evaluate the performance.

本文提出了一种基于高斯拉普拉斯算子 (LoG) 的关键点检测方法。与尺度不变特征变换 (SIFT) 中使用的基于高斯差分 (DoG) 的关键点检测方法相比, 我们重点研究 LoG 算子及其高阶导数。我们通过高阶 DoG (HDoG) 和高阶 LoG (HLoG) 之间的数学类比, 并结合实验结果, 证明了本文提出的基于 HLoG 的关键点检测方法的有效性。HLoG 的性能通过四种不同的测试进行评估: i) 对不同变换下图像中检测到的关键点的可重复性测试; ii) 图像检索; iii) 全景图拼接; iv) 三维重建。本文提出的 HLoG 方法的性能与 HDoG 相当, 并且 HLoG 与 HDoG 的结合在各种关键点相关的计算机视觉问题上取得了显著的改进。

综上所述, 我们提出了一种基于高阶空间导数的关键点检测方法 HLoG。我们通过四个不同的实验评估证明了该方法的有效性。首先, 使用一小组已知单应性信息的图像进行可重复性测试。其次, 使用包含大量图像的数据集进行图像检索测试。第三, 进行全景拼接测试, 以验证该方法在图像配准问题中的有效性。最后, 我们进行了三维重建测试, 并验证了该方法能够生成更多三维点。我们成功地证明了, 所提出的基于 HLoG 的关键点检测方法与 HDoG 结合使用, 在解决各种计算机视觉问题时具有显著优势。我们认为, HDoG 与 HLoG 融合的性能提升源于基础算子 (即 DoG 与 LoG) 之间的差异。HDoG 和 HLoG 之间的差异在每个推导阶数上都会累积, 并且在考虑多个推导阶数时会变得显著。我们将进一步评估 HDoG 和 HLoG, 以了解它们的特性和最佳参数设置。我们正在优化 HLoG 滤波器, 并尝试应用一种归一化技术, 以稳定滤波器对高频分量的响应。我们还将 HLoG 应用于更多计算机视觉问题, 例如二维或三维对齐, 并评估其性能。

The low-rank matrix approximation (LRMA) is an efficient image denoising method to reduce additive Gaussian noise. However, the existing low-rank matrix approximation does not perform well in terms of Rician noise removal for magnetic resonance imaging (MRI). To this end, we propose a novel MR image denoising approach based on the extended difference of Gaussian (DoG) filter and nonlocal low-rank regularization. In the proposed method, a novel nonlocal self-similarity evaluation with the tight frame is exploited to improve the patch matching. To remove the Rician noise and preserve the edge details, the extended DoG filter is exploited to the nonlocal low-rank regularization model. The experimental results demonstrate that the proposed method can preserve more edge and fine structures while removing noise in MR image as compared with some of the existing methods.

In Figs. 2, 3, and 4, we carry out the visual comparisons of the denoised MR images by the different denoising methods. It is observed that the proposed DoG-WNNM is very similar to the true image in visual perception, in which some edges and fine structures are better preserved, and it significantly improves the image sharpness. The reason is that our method sufficiently exploits the NSS structure of image and the DoG filter operation in each group patch procedure. To further demonstrate the effectiveness of the DoG-WNNM, we compare the difference between the original image and denoising results in the third row of Figs. 2, 3, and 4. It is evident that all of denoising methods demonstrate remarkable performance in removing noise. The NLM can be effective to remove the noise, but it loses the parts of texture at edge regions, which enables blurring of the tissue structures of interest. By contrast, the images produced by DoG-WNNM exhibit less texture at the third row, and better preserve the edges and texture details. In Figs. 5 and 6, we further address the MR image denoising with the real data. Compared with the other competing algorithms, the output of the DoG-WNNM is visually more pleasant. Therefore, the proposed method enhances the edges and fine structures of images and effectively reduces the noise.

This paper proposes a novel denoising framework based on the DoG filter and nonlocal low-rank model, which improves the quality of denoised image. To demonstrate the effectiveness of the DoG-WNNM for MR image denoising, we combine the novel NSS prior of image and the extended DoG filter together to construct the low-rank regularization model. Moreover, the equivalent model is also

proposed to solve the WNNM regularization model. Experimental results with different noise levels demonstrate the advantages of the DoG-WNNM. Compared with the existing denoising methods, the proposed method has better performance in preserving the edge structures and detailed textures of the image. Theoretical analysis about the Rician bias correction is planned as part of a future work.

低秩矩阵近似 (LRMA) 是一种有效的图像去噪方法,可以降低加性高斯噪声。然而,现有的低秩矩阵近似在磁共振成像 (MRI) 的莱斯噪声去除方面效果不佳。为此,我们提出了一种基于扩展高斯差分 (DoG) 滤波器和非局部低秩正则化的磁共振图像去噪方法。在该方法中,利用一种新的紧框架非局部自相似性评估来改进块匹配。为了去除莱斯噪声并保留边缘细节,我们将扩展的 DoG 滤波器应用于非局部低秩正则化模型。实验结果表明,与一些现有方法相比,该方法在去除磁共振图像噪声的同时,能够保留更多的边缘和精细结构。

在图 2、3 和 4 中,我们对不同去噪方法对 MRI 图像进行了视觉比较。结果表明,所提出的 DoG-WNNM 在视觉感知上与真实图像非常相似,部分边缘和精细结构得到了更好的保留,并且显著提高了图像的清晰度。这是因为,我们的方法充分利用了图像的 NSS 结构以及每个组块处理过程中的 DoG 滤波器操作。为了进一步证明 DoG-WNNM 的有效性,我们在图 2、3 和 4 的第三行比较了原始图像与去噪结果之间的差异。显然,所有去噪方法在去噪方面都表现出了显著的效果。NLM 可以有效地去除噪声,但它会丢失边缘区域的纹理部分,从而导致感兴趣的组织结构变得模糊。相比之下,DoG-WNNM 生成的图像在第三行纹理较少,边缘和纹理细节保留得更好。在图 5 和图 6 中,我们进一步利用真实数据对磁共振图像进行去噪处理。与其他竞争算法相比,DoG-WNNM 的输出在视觉上更加美观。因此,本文提出的方法增强了图像的边缘和精细结构,并有效地降低了噪声。

本文提出了一种基于 DoG 滤波器和非局部低秩模型的新型去噪框架,提高了去噪图像的质量。为了证明 DoG-WNNM 对磁共振图像去噪的有效性,我们将新的图像 NSS 先验与扩展的 DoG 滤波器相结合,构建了低秩正则化模型。此外,我们还提出了等效模型来求解 WNNM 正则化模型。不同噪声水平下的实验结果证明了 DoG-WNNM 的优势。与现有的去噪方法相比,所提方法在保留图像边缘结构和细节纹理方面具有更好的效果。关于莱斯偏差校正的理论分析计划在未来的工作中展开。

We develop a new edge detection algorithm that addresses two important issues in this long-standing vision problem: (1) holistic image training and prediction; and

(2) multi-scale and multi-level feature learning. Our proposed method, holistically-nested edge detection (HED), performs image-to-image prediction by means of a deep learning model that leverages fully convolutional neural networks and deeply-supervised nets. HED automatically learns rich hierarchical representations (guided by deep supervision on side responses) that are important in order to resolve the challenging ambiguity in edge and object boundary detection. We significantly advance the state-of-the-art on the BSD500 dataset (ODS F-score of 0.782) and the NYU Depth dataset (ODS F-score of 0.746), and do so with an improved speed (0.4s per image) that is orders of magnitude faster than some recent CNN-based edge detection algorithms.

In this paper, we have developed a new convolutional-neural-network-based edge detection system that demonstrates state-of-the-art performance on natural images at a speed of practical relevance (e.g., 0.4 seconds using GPU and 12 seconds using CPU). Our algorithm builds on top of the ideas of fully convolutional neural networks and deeply-supervised nets. We also initialize our network structure and parameters by adopting a pre-trained trimmed VG-GNet. Our method shows promising results in performing image-to-image learning by combining multi-scale and multi-level visual responses, even though explicit contextual and high-level information has not been enforced.

我们开发了一种新的边缘检测算法，解决了这一长期存在的视觉问题中的两个重要问题：(1) 整体图像训练和预测；(2) 多尺度和多层次的特征学习。我们提出的方法，即整体嵌套边缘检测 (HED)，通过一个利用全卷积神经网络和深度监督网络的深度学习模型进行图像到图像的预测。HED 自动学习丰富的层次化表示（由对侧向响应的深度监督引导），这对于解决边缘和物体边界检测中棘手的模糊性问题至关重要。我们在 BSD500 数据集（ODS F 值 0.782）和 NYU 深度数据集（ODS F 值 0.746）上显著提升了当前最佳水平，并且速度有所提升（每幅图像 0.4 秒），比一些近期基于 CNN 的边缘检测算法快了几个数量级。

本文开发了一种基于卷积神经网络的新型边缘检测系统，该系统在自然图像上展现了最佳性能，并且速度达到了实用要求（例如，使用 GPU 为 0.4 秒，使用 CPU 为 12 秒）。我们的算法建立在全卷积神经网络和深度监督网络的思想之上。我们还采用预先训练好的修剪型 VG-GNet 初始化了网络结构

和参数。我们的方法通过结合多尺度和多层次的视觉响应，在执行图像到图像的学习方面显示出良好的效果，尽管没有强制执行明确的上下文和高级信息。

The encoder-decoder framework is state-of-the-art for offline semantic image segmentation. Since the rise in autonomous systems, real-time computation is increasingly desirable. In this paper, we introduce fast segmentation convolutional neural network (Fast-SCNN), an above real-time semantic segmentation model on high resolution image data ($1024 \times 2048\text{px}$) suited to efficient computation on embedded devices with low memory. Building on existing two-branch methods for fast segmentation, we introduce our ‘learning to downsample’ module which computes low-level features for multiple resolution branches simultaneously. Our network combines spatial detail at high resolution with deep features extracted at lower resolution, yielding an accuracy of 68.0% mean intersection over union at 123.5 frames per second on Cityscapes. We also show that large scale pre-training is unnecessary. We thoroughly validate our metric in experiments with ImageNet pre-training and the coarse labeled data of Cityscapes. Finally, we show even faster computation with competitive results on sub-sampled inputs, without any network modifications.

Fast semantic segmentation is particularly important in real-time applications, where input is to be parsed quickly to facilitate responsive interactivity with the environment. Due to the increasing interest in autonomous systems and robotics, it is therefore evident that the research into real-time semantic segmentation has recently enjoyed significant gain in popularity [21, 34, 17, 25, 36, 20]. We emphasize, faster than real-time performance is in fact often necessary, since semantic labeling is usually employed only as preprocessing step of other time-critical tasks. Furthermore, real-time semantic segmentation on embedded devices (without access to powerful GPUs) may enable many additional applications: • a larger receptive field is important to learn complex correlations among object classes (i.e. global context), • spatial detail in images is necessary to preserve object boundaries, and • specific designs are needed to balance speed and accuracy (rather than re-targeting classification DCNNs). Specifically in the two-branch networks, a deeper branch is employed at low resolution to capture global context, while a shallow branch is setup to learn spatial details at full input resolution. The final semantic segmentation result is then provided by merging the two. Importantly since the computational cost of

deeper networks is overcome with a small input size, and execution on full resolution is only employed for few layers, real-time performance is possible on modern GPUs. In contrast to the encoder-decoder framework, the initial convolutions at different resolutions are not shared in the two-branch approach. Here it is worth noting, the guided upsampling network (GUN) [17] and the image cascade network (ICNet) [36] only share the weights among the first few layers, but not the computation. In this work we propose fast segmentation convolutional neural network Fast-SCNN, an above real-time semantic segmentation algorithm merging the two-branch setup of prior art [21, 34, 17, 36], with the classical encoder-decoder framework [29, 2] (Figure 1). Building on the observation that initial DCNN layers extract low-level features [35, 19], we share the computations of the initial layers in the two-branch approach. We call this technique learning to down-sample. The effect is similar to a skip connection in the encoder-decoder model, but the skip is only employed once to retain runtime efficiency, and the module is kept shallow to ensure validity of feature sharing. Finally, our Fast-SCNN adopts efficient depthwise separable convolutions [30, 10], and inverse residual blocks [28]. Applied on Cityscapes [6], Fast-SCNN yields a mean intersection over union (mIoU) of 68.0% at 123.5 frames per second (fps) on a modern GPU (Nvidia Titan Xp (Pascal)) using full (1024×2048px) resolution, which is twice as fast as prior art i.e. BiSeNet (71.4% mIoU)[34]. While we use 1.11 million parameters, most offline segmentation methods (e.g. DeepLab [4] and PSPNet [37]), and some real-time algorithms (e.g. GUN [17] and ICNet [36]) require much more than this. The model capacity of Fast-SCNN is kept specifically low. The reason is two-fold: (i) lower memory enables execution on embedded devices, and (ii) better generalisation is expected. In particular, pre-training on ImageNet [27] is frequently advised to boost accuracy and generality [37]. In our work, we study the effect of pre-training on the low capacity Fast-SCNN. Contradicting the trend of high-capacity networks, we find that results only insignificantly improve with pre-training or additional coarsely labeled training data (+0.5% mIoU on Cityscapes [6]). In summary our contributions are: 1. We propose Fast-SCNN, a competitive (68.0%) and above real-time semantic segmentation algorithm (123.5 fps) for high resolution images (1024 × 2048px). 1. We adapt the skip connection, popular in offline DCNNs, and propose a shallow learning to downsample module for fast and efficient multi-branch low-level feature extraction. 3. We specifically design Fast-SCNN to be of low capacity, and we empirically

validate that running training for more epochs is equivalently successful to pre-training with ImageNet or training with additional coarse data in our small capacity network. Moreover, we employ Fast-SCNN to subsampled input data, achieving state-of-the-art performance without the need for redesigning our network.

Fast-SCNN is inspired by the two-branch architectures [21, 34, 17] and encoder-decoder networks with skip connections [29, 26]. Noting that early layers commonly extract low-level features. We reinterpret skip connections as a learning to downsample module, enabling us to merge the key ideas of both frameworks, and allowing us to build a fast semantic segmentation model. Figure 1 and Table 1 present the layout of Fast-SCNN. In the following we discuss our motivation and describe our building blocks in more detail.

3.1. Motivation

Current state-of-the-art semantic segmentation methods that run in real-time are based on networks with two branches, each operating on a different resolution level [21, 34, 17]. They learn global information from low-resolution versions of the input image, and shallow networks at full input resolution are employed to refine the precision of the segmentation results. Since input resolution and network depth are main factors for runtime, these two-branch approaches allow for real-time computation. It is well known that the first few layers of DCNNs extract the low-level features, such as edges and corners [35, 19]. Therefore, rather than employing a two-branch approach with separate computation, we introduce learning to downsample, which shares feature computation between the low and high-level branch in a shallow network block.

3.2. Network Architecture

Our Fast-SCNN uses a learning to downsample module, a coarse global feature extractor, a feature fusion module and a standard classifier. All modules are built using depth-wise separable convolution, which has become a key building block for many efficient DCNN architectures [5, 10, 21].

3.2.1 Learning to Downsample

In our learning to downsample module, we employ three layers. Only three layers are employed to ensure low-level feature sharing is valid, and efficiently implemented. The first layer is a standard convolutional layer (Conv2D) and the remaining two layers are depthwise separable convolutional layers (DSConv). Here we emphasize, although DSConv is computationally more efficient, we employ Conv2D since the input image only has three channels, making DSConv’s computational benefit insignificant at this stage. All three layers in our learning to downsample module use stride 2, followed by batch normalization [12] and ReLU.

The spatial kernel size of the convolutional and depthwise layers is 3×3 . Following [5, 28, 21], we omit the nonlinearity between depthwise and pointwise convolutions.

3.2.2 Global Feature Extractor The global feature extractor module is aimed at capturing the global context for image segmentation. In contrast to common two-branch methods which operate on low-resolution versions of the input image, our module directly takes the output of the learning to downsample module (which is at 1/8-resolution of the original input). The detailed structure of the module is shown in Table 1. We use efficient bottleneck residual block introduced by MobileNet-V2 [28] (Table 2). In particular, we employ residual connection for the bottleneck residual blocks when the input and output are of the same size. Our bottleneck block uses an efficient depthwise separable convolution, resulting in less number of parameters and floating point operations. Also, a pyramid pooling module (PPM) [37] is added at the end to aggregate the different-region-based context information.

3.2.3 Feature Fusion Module Similar to ICNet [36] and ContextNet [21] we prefer simple addition of the features to ensure efficiency. Alternatively, more sophisticated feature fusion modules (e.g. [34]) could be employed at the cost of runtime performance, to reach better accuracy. The detail of the feature fusion module is shown in Table 3.

3.2.4 Classifier In the classifier we employ two depthwise separable convolutions (DSConv) and one pointwise convolution (Conv2D). We found that adding few layers after the feature fusion module boosts the accuracy. The details of the classifier module is shown in the Table 1. Softmax is used during training, since gradient decent is employed. During inference we may substitute costly softmax computations with argmax, since both functions are monotonically increasing. We denote this option as Fast-SCNN cls (classification). On the other hand, if a standard DCNN based probabilistic model is desired, softmax is used, denoted as Fast-SCNN prob (probability).

3.3. Comparison with Prior Art Our model is inspired by the two-branch framework, and incorporates ideas of encoder-decoder methods (Figure 2).

3.3.1 Relation with Two-branch Models The state-of-the-art real-time models (ContextNet [21], BiSeNet [34] and GUN [17]) use two-branch networks. Our learning to downsample module is equivalent to their spatial path, as it is shallow, learns from full resolution, and is used in the feature fusion module (Figure 1). Our global feature extractor module is equivalent to the deeper low-resolution branch of such approaches. In contrast, our global feature extractor shares its computation of the first few layers with the learning to downsample module. By sharing the layers we

not only reduce computational complexity of feature extraction, but we also reduce the required input size as Fast-SCNN uses 1/8-resolution instead of 1/4-resolution for global feature extraction.

3.3.2 Relation with Encoder-Decoder Models

Proposed Fast-SCNN can be viewed as a special case of an encoder-decoder framework, such as FCN [29] or U-Net [26]. However, unlike the multiple skip connections in FCN and the dense skip connections in U-Net, Fast-SCNN only employs a single skip connection to reduce computations as well as memory. In correspondence with [35], who advocate that features are shared only at early layers in DCNNs, we position our skip connection early in our network. In contrast, prior art typically employ deeper modules at each resolution, before skip connections are applied.

4 Experiments

We evaluated our proposed fast segmentation convolutional neural network (Fast-SCNN) on the validation set of the Cityscapes dataset [6], and report its performance on the Cityscapes test set, i.e. the Cityscapes benchmark server.

4.1. Implementation Details

Implementation detail is as important as theory when it comes to efficient DCNNs. Hence, we carefully describe our setup here. We conduct experiments on the TensorFlow machine learning platform using Python. Our experiments are executed on a workstation with either Nvidia Titan X (Maxwell) or Nvidia Titan Xp (Pascal) GPU, with CUDA 9.0 and CuDNN v7. Runtime evaluation is performed in a single CPU thread and one GPU to measure the forward inference time. We use 100 frames for burn-in and report average of 100 frames for the frames per second (fps) measurement. We use stochastic gradient descent (SGD) with momentum 0.9 and batch-size 12. Inspired by [4, 37, 10] we use ‘poly’ learning rate with the base one as 0.045 and power as 0.9. Similar to MobileNet-V2 we found that ℓ_2 regularization is not necessary on depthwise convolutions, for other layers ℓ_2 is 0.00004. Since training data for semantic segmentation is limited, we apply various data augmentation techniques: random resizing between 0.5 to 2, translation/crop, horizontal flip, color channels noise and brightness. Our model is trained with cross-entropy loss. We found that auxiliary losses at the end of learning to down-sample and the global feature extraction modules with 0.4 weights are beneficial. Batch normalization [12] is used before every non-linear function. Dropout is used only on the last layer, just before the softmax layer. Contrary to MobileNet [10] and ContextNet [21], we found that Fast-SCNN trains faster with ReLU and achieves slightly better accuracy than ReLU6, even with the depthwise separable convolutions that we use throughout our model. We found that the performance of DCNNs can be improved

by training for higher number of iterations, hence we train our model for 1,000 epochs unless otherwise stated, using the Cityscapes dataset 6. It is worth noting here, Fast-SCNN's capacity is deliberately very low, as we employ 1.11 million parameters. Later we show that aggressive data augmentation techniques make overfitting unlikely.

编码器-解码器框架是离线语义图像分割的最新成果。随着自主系统的兴起,实时计算的需求日益增长。本文介绍了一种快速分割卷积神经网络 (Fast-SCNN),这是一种基于高分辨率图像数据 ($1024 \times 2048\text{px}$) 的超实时语义分割模型,适用于低内存嵌入式设备进行高效计算。在现有的双分支快速分割方法的基础上,我们引入了“学习下采样”模块,该模块可同时计算多个分辨率分支的低级特征。我们的网络将高分辨率的空间细节与低分辨率提取的深度特征相结合,在 Cityscapes 数据集上以每秒 123.5 帧的速度实现了 68.0% 的平均交并比准确率。我们还证明了大规模预训练的必要性。我们在 ImageNet 预训练数据集和 Cityscapes 的粗标注数据上进行了实验,彻底验证了我们的指标。最终,我们展示了更快的计算速度,并在子采样输入上取得了具有竞争力的结果,而无需任何网络修改。

快速语义分割在实时应用中尤为重要,因为这些应用中需要快速解析输入,以便与环境进行响应式交互。由于人们对自主系统和机器人技术的兴趣日益浓厚,实时语义分割的研究近年来显著普及 [21, 34, 17, 25, 36, 20]。我们强调,实际上,更快的性能往往是必要的,因为语义标注通常仅用作其他时间关键型任务的预处理步骤。此外,在嵌入式设备上实现实时语义分割(无需使用强大的 GPU)可以实现许多其他功能:

- 更大的感受野对于学习对象类别之间的复杂关联(即全局上下文)至关重要;
- 图像中的空间细节对于保留对象边界至关重要;
- 需要特定的设计来平衡速度和准确性(而不是重新定位分类 DCNN)。

具体来说,在双分支网络中,一个较深的分支用于在低分辨率下捕捉全局上下文,而一个较浅的分支用于在全输入分辨率下学习空间细节。最终的语义分割结果由两者合并而成。重要的是,由于较小输入规模可以克服较深网络的计算成本,并且仅对少数层采用全分辨率执行,因此在现代 GPU 上实现实时性能是可能的。与编码器-解码器框架不同,在双分支方法中,不同分辨率的初始卷积层并不共享。需要注意的是,引导上采样网络 (GUN) 17 和图像级联网络 (ICNet) [36] 仅在前几层之间共享权重,但不共享计算。本文提出了一种快速分割卷积神经网络 Fast-SCNN,这是一种超实时语义分割算法,它将现有技术 [21, 34, 17, 36] 的双分支设置与经典的编码器-解码器框架 [29, 2] 相结

合（图 1）。基于初始 DCNN 层提取低级特征 [35, 19] 的观察，我们在双分支方法中共享初始层的计算。我们将这种技术称为学习下采样。其效果类似于编码器-解码器模型中的跳过连接，但跳过连接仅使用一次以保持运行时效率，并且模块保持较浅以确保特征共享的有效性。最后，我们的 Fast-SCNN 采用高效的深度可分离卷积 [30, 10] 和逆残差块 [28]。在 Cityscapes 6 数据集上应用 Fast-SCNN 后，在现代 GPU（Nvidia Titan Xp (Pascal)）上使用全分辨率（1024×2048px）时，以 123.5 帧/秒 (fps) 的速度实现了 68.0% 的平均交并比 (mIoU)，速度是现有技术 BiSeNet (71.4% mIoU) [34] 的两倍。虽然我们使用了 111 万个参数，但大多数离线分割方法（例如 DeepLab 4 和 PSPNet [37]）以及一些实时算法（例如 GUN 17 和 ICNet [36]）所需的参数远超此范围。Fast-SCNN 的模型容量被控制得特别低。原因有两个：(i) 较低的内存使其能够在嵌入式设备上执行；(ii) 期望获得更好的泛化能力。特别是，通常建议在 ImageNet [27] 上进行预训练以提高准确率和泛化能力 [37]。在我们的研究中，我们研究了预训练对低容量 Fast-SCNN 的影响。与高容量网络的趋势相反，我们发现，预训练或额外的粗标记训练数据对结果的提升并不显著（在 Cityscapes 6 数据集上 mIoU 提高了 0.5% 6）。总结一下，我们的贡献如下：

1. 我们提出了 Fast-SCNN，这是一款针对高分辨率图像（1024 × 2048px）的实时语义分割算法，具有竞争力（68.0%），且达到 123.5 fps。

我们采用了在离线深度神经网络 (DCNN) 中流行的跳跃连接 (skip connection)，并提出了一种浅层学习下采样模块，以实现快速高效的多分支低级特征提取。

我们专门设计了低容量的 Fast-SCNN，并通过经验验证，在我们的小容量网络中，进行更多轮次的训练与使用 ImageNet 进行预训练或使用额外的粗略数据进行训练的效果相同。此外，我们使用 Fast-SCNN 对输入数据进行下采样，无需重新设计网络即可实现最佳性能。

Fast-SCNN 的灵感来源于双分支架构 [21, 34, 17] 和带有跳跃连接的编码器-解码器网络 [29, 26]。我们注意到，早期层通常提取低级特征。我们将跳跃连接重新解释为学习下采样模块，这使我们能够融合两个框架的关键思想，并构建一个快速的语义分割模型。图 1 和表 1 展示了 Fast-SCNN 的布局。接下来，我们将讨论我们的动机，并更详细地描述我们的构建模块。

3.1. 动机

目前最先进的实时语义分割方法基于具有两个分支的网络，每个分支在不同的分辨率级别上运行 [21, 34, 17]。它们从输入图像的低分辨率版本中学习全局信息，并使用全输入分辨率的浅层网络来提高分割结果的精度。由于输入分辨率

和网络深度是影响运行时间的主要因素，这些双分支方法可以实现实时计算。众所周知，深度卷积神经网络 (DCNN) 的前几层提取低级特征，例如边和角 [35, 19]。因此，我们并未采用单独计算的双分支方法，而是引入了下采样学习 (Learning to Downsample)，该方法在浅层网络模块中，在低级和高级分支之间共享特征计算。

3.2. 网络架构 我们的 Fast-SCNN 使用一个下采样学习模块、一个粗略全局特征提取器、一个特征融合模块和一个标准分类器。所有模块均采用深度可分离卷积构建，这已成为许多高效 DCNN 架构的关键构建模块 [5, 10, 21]。

3.2.1 下采样学习 在我们的下采样学习模块中，我们采用了三层结构。为了确保低级特征共享的有效性和高效性，我们仅使用了三层。第一层是标准卷积层 (Conv2D)，其余两层是深度可分离卷积层 (DSConv)。需要强调的是，尽管 DSConv 的计算效率更高，但由于输入图像只有三个通道，我们采用了 Conv2D，因此 DSConv 的计算优势在现阶段并不显著。我们的学习下采样模块中的所有三层都采用步长 2，并随后进行批量归一化 [12] 和 ReLU 操作。卷积层和深度层的空间核大小均为 3×3 。遵循 [5, 28, 21] 的经验，我们省略了深度卷积和逐点卷积之间的非线性关系。

3.2.2 全局特征提取器 全局特征提取器模块旨在捕捉图像分割的全局上下文信息。与常见的双分支方法（它们对输入图像的低分辨率版本进行操作）不同，我们的模块直接获取学习下采样模块的输出（其分辨率为原始输入的 $1/8$ 倍）。该模块的详细结构如表 1 所示。我们使用了 MobileNet-V2 [28] 中引入的高效瓶颈残差块（表 2）。具体而言，当输入和输出大小相同时，我们对瓶颈残差块采用残差连接。我们的瓶颈块使用高效的深度可分离卷积，从而减少了参数数量和浮点运算次数。此外，在模块末尾添加了一个金字塔池化模块 (PPM) [37]，用于聚合基于不同区域的上下文信息。

3.2.3 特征融合模块 与 ICNet [36] 和 ContextNet [21] 类似，我们倾向于简单地添加特征以确保效率。或者，也可以使用更复杂的特征融合模块（例如 [34]），但会牺牲运行时性能，以达到更高的准确率。特征融合模块的细节如表 3 所示。

3.2.4 分类器 在分类器中，我们采用了两个深度可分离卷积 (DSConv) 和一个逐点卷积 (Conv2D)。我们发现，在特征融合模块后添加少量层可以提高准确率。分类器模块的细节如表 1 所示。训练过程中使用 Softmax，因为其中采用了梯度下降。在推理过程中，我们可能会用 argmax 代替计算成本高昂的 softmax，因为这两个函数都是单调递增的。我们将此选项称为 Fast-SCNN cls（分类）。另一方面，如果需要基于标准 DCNN 的概率模型，则使用 softmax，表示为 Fast-SCNN prob（概率）。

3.3. 与现有技术的比较 我们的模型受到双分支框架的启发，并融入了编码器-解码器方法的思想（图 2）。

3.3.1 与双分支模

型的关系 最先进的实时模型（ContextNet 21、 BiSeNet [34] 和 GUN 17）都使用双分支网络。我们的 学习下采样模块相当于它们的空间路径，因为它比较浅，从全分辨率学习，并且用于特征融合模块（图 1）。 我们的全局特征提取器模块相当于此类方法中更深的低分辨率分支。相比之下，我们的全局特征提取器与学习下采样模块共享了前几层的计算。通过共享这些层，我们不仅降低了特征提取的计算复杂度，而且还减少了所需的输入大小，因为 Fast-SCNN 使用 1/8 分辨率而不是 1/4 分辨率进行全局特征提取。

3.3.2 与编码器-解码器模型的关系 提出的 Fast-SCNN 可以被视为编码器-解码器框架的一个特例，例如 FCN [29] 或 U-Net [26]。然而，与 FCN 中的多个跳跃连接和 U-Net 中的密集跳跃连接不同，Fast-SCNN 仅使用单个跳跃连接来减少计算量和内存占用。与 [35] 主张特征仅在深度卷积神经网络 (DCNN) 的早期层共享的观点一致，我们将跳跃连接放置在网络的早期层。相比之下，现有技术通常在每个分辨率下都采用更深的模块，然后再应用跳跃连接。

4.实验 我们在 Cityscapes 数据集 6 的验证集上评估了我们提出的快速分割卷积神经网络 (Fast-SCNN)，并在 Cityscapes 测试集（即 Cityscapes 基准服务器）上报告了其性能。

4.1. 实现细节 对于高效的深度卷积神经网络 (DCNN) 而言，实现细节与理论同样重要。因此，我们在此仔细描述我们的设置。我们使用 Python 在 TensorFlow 机器学习平台上进行实验。我们的实验在配备 Nvidia Titan X (Maxwell) 或 Nvidia Titan Xp (Pascal) GPU、CUDA 9.0 和 CuDNN v7 的工作站上执行。运行时评估在单个 CPU 线程和一个 GPU 上进行，以测量前向推理时间。我们使用 100 帧进行老化测试，并报告 每秒帧数 (fps) 测量的平均值。 我们使用动量为 0.9、批量大小为 12 的随机梯度下降 (SGD)。受 [4, 37, 10] 的启发，我们使用 “poly”学习率，基数为 0.045，幂为 0.9。与 MobileNet-V2 类似，我们发现深度卷积不需要 L2 正则化，对于其他层，L2 为 0.00004。由于语义分割的训练数据有限，我们应用了各种数据增强技术：在 0.5 到 2 之间随机调整大小、平移/裁剪、水平翻转、颜色通道噪声和亮度增强。我们的模型采用交叉熵损失进行训练。我们发现，在学习下采样结束时，辅助损失和全局特征提取模块的 0.4 权重是有益的。 每个非线性函数之前都使用了批量归一化 12。仅在最后一层（紧邻 softmax 层）使用了 Dropout。与 MobileNet 10 和 ContextNet 21 相反，我们发现 Fast-SCNN 使用 ReLU 训练速度更快，并且准确率略高于 ReLU6， 即使我们在整个模型中使用了深度可分离卷积也是如此。 我们发现，通过增加训练迭代次数可以提高深度卷积神经网络 (DCNN) 的性能，因此，除非另有说明，否则我们使用 Cityscapes 数据集 6 对模型进行 1,000 次迭代训练。需要注意的是， Fast-

SCNN 的容量刻意很低，因为我们使用了 111 万个参数。之后我们将展示，激进的数据增强技术可以降低过拟合的可能性。

我们提出了一个快速分割网络，用于实现上述实时场景理解。分担多分支网络的计算成本可提高运行效率。实验表明，我们的跳跃连接有助于恢复空间细节。我们还证明，如果训练时间足够长，对于低容量网络而言，无需在额外的辅助任务上对模型进行大规模预训练。

Simple Online and Realtime Tracking (SORT) is a pragmatic approach to multiple object tracking with a focus on simple, effective algorithms. In this paper, we integrate appearance information to improve the performance of SORT. Due to this extension we are able to track objects through longer periods of occlusions, effectively reducing the number of identity switches. In spirit of the original framework we place much of the computational complexity into an offline pre-training stage where we learn a deep association metric on a large-scale person re-identification dataset. During online application, we establish measurement-to-track associations using nearest neighbor queries in visual appearance space. Experimental evaluation shows that our extensions reduce the number of identity switches by 45%, achieving overall competitive performance at high frame rates.

Due to recent progress in object detection, tracking-by-detection has become the leading paradigm in multiple object tracking. Within this paradigm, object trajectories are usually found in a global optimization problem that processes entire video batches at once. For example, flow network formulations [1, 2, 3] and probabilistic graphical models [4, 5, 6, 7] have become popular frameworks of this type. However, due to batch processing, these methods are not applicable in online scenarios where a target identity must be available at each time step. More traditional methods are Multiple Hypothesis Tracking (MHT) [8] and the Joint Probabilistic Data Association Filter (JPDAF) [9]. These methods perform data association on a frame-by-frame basis. In the JPDAF, a single state hypothesis is generated by weighting individual measurements by their association likelihoods. In MHT, all possible hypotheses are tracked, but pruning schemes must be applied for computational tractability. Both methods have recently been revisited in a tracking-by-detection scenario [10, 11] and shown promising results. However, the performance of these methods comes at increased computational and implementation complexity. Simple online and realtime tracking (SORT) [12] is a much simpler framework that performs Kalman filtering in image

space and frame-by-frame data association using the Hungarian method with an association metric that measures bounding box overlap. This simple approach achieves favorable performance at high frame rates. On the MOT challenge dataset 13, SORT with a state-of-the-art people detector 14 ranks on average higher than MHT on standard detections. This not only underlines the influence of object detector performance on overall tracking results, but is also an important insight from a practitioners point of view. While achieving overall good performance in terms of tracking precision and accuracy, SORT returns a relatively high number of identity switches. This is, because the employed association metric is only accurate when state estimation uncertainty is low. Therefore, SORT has a deficiency in tracking through occlusions as they typically appear in frontal-view camera scenes. We overcome this issue by replacing the association metric with a more informed metric that combines motion and appearance information. In particular, we apply a convolutional neural network (CNN) that has been trained to discriminate pedestrians on a large-scale person re-identification dataset. Through integration of this network we increase robustness against misses and occlusions while keeping the system easy to implement, efficient, and applicable to online scenarios. Our code and a pre-trained CNN model are made publicly available to facilitate research experimentation and practical application development along with a script that can be used to generate features.

EXPERIMENTS We assess the performance of our tracker on the MOT16 benchmark 15. This benchmark evaluates tracking performance on seven challenging test sequences, including frontal-view scenes with moving camera as well as top-down surveillance setups. As input to our tracker we rely on detections provided by Yu et al. 16. They have trained a Faster RCNN on a collection of public and private datasets to provide excellent performance. For a fair comparison, we have re-run SORT on the same detections. Evaluation on test sequences were carried out using $\lambda = 0$ and $A_{max} = 30$ frames. As in 16, detections have been thresholded at a confidence score of 0.3. The remaining parameters of our method have been found on separate training sequences which are provided by the benchmark. Evaluation is carried out according to the following metrics:

- Multi-object tracking accuracy (MOTA): Summary of overall tracking accuracy in terms of false positives, false negatives and identity switches 23.
- Multi-object tracking precision (MOTP): Summary of overall tracking precision in terms of bounding box overlap between

ground-truth and reported location 23. • Mostly tracked (MT): Percentage of ground-truth tracks that have the same label for at least 80% of their life span. • Mostly lost (ML): Percentage of ground-truth tracks that are tracked for at most 20% of their life span. • Identity switches (ID): Number of times the reported identity of a ground-truth track changes. • Fragmentation (FM): Number of times a track is interrupted by a missing detection. The results of our evaluation are shown in Table 2. Our adaptations successfully reduce the number of identity switches. In comparison to SORT, ID switches reduce from 1423 to 781. This is a decrease of approximately 45%. At the same time, track fragmentation increase slightly due to maintaining object identities through occlusions and misses. We also see a significant increase in number of mostly tracked objects and a decrease of mostly lost objects. Overall, due to integration of appearance information we successfully maintain identities through longer occlusions. This can also be seen by qualitative analysis of the tracking output that we provide in the supplementary material. An exemplary output of our tracker is shown in Figure 1. Our method is also a strong competitor to other online tracking frameworks. In particular, our approach returns the fewest number of identity switches of all online methods while maintaining competitive MOTA scores, track fragmentations, and false negatives. The reported tracking accuracy is mostly impaired by a larger number of false positives. Given their overall impact on the MOTA score, applying a larger confidence threshold to the detections can potentially increase the reported performance of our algorithm by a large margin. However, visual inspection of the tracking output shows that these false positives are mostly generated from sporadic detector responses at static scene geometry. Due to our relatively large maximum allowed track age, these are more commonly joined to object trajectories. At the same time, we did not observe tracks jumping between false alarms frequently. Instead, the tracker commonly generated relatively stable, stationary tracks at the reported object location. Our implementation runs at approximately 20 Hz with roughly half of the time spent on feature generation. Therefore, given a modern GPU, the system remains computationally efficient and operates at real time.

CONCLUSION We have presented an extension to SORT that incorporates appearance information through a pre-trained association metric. Due to this extension, we are able to track through longer periods of occlusion, making SORT a strong

competitor to state-of-the-art online tracking algorithms. Yet, the algorithm remains simple to implement and runs in real time.

简单在线实时跟踪 (SORT) 是一种实用的多目标跟踪方法, 注重简单、高效的算法。本文中, 我们整合了外观信息来提升 SORT 的性能。得益于此扩展, 我们能够在较长的遮挡周期内跟踪目标, 从而有效减少身份切换的次数。秉承原始框架的理念, 我们将大部分计算复杂度放在离线预训练阶段, 在该阶段, 我们基于大规模行人重识别数据集学习深度关联度量。在线应用中, 我们使用视觉外观空间中的最近邻查询建立测量到跟踪的关联。实验评估表明, 我们的扩展将身份切换次数减少了 45%, 在高帧率下实现了具有竞争力的整体性能。

由于近年来目标检测技术的进步, 逐检测跟踪已成为多目标跟踪的主流范式。在该范式中, 目标轨迹通常存在于一次性处理整个视频批次的全局优化问题中。例如, 流网络公式 [1, 2, 3] 和概率图模型 [4, 5, 6, 7] 已成为此类流行的框架。然而, 由于批处理的原因, 这些方法并不适用于在线场景, 因为在线场景中每个时间步都必须有目标身份。更传统的方法是多假设跟踪 (MHT) 8 和联合概率数据关联滤波器 (JPDAF) 9。这些方法逐帧执行数据关联。在 JPDAF 中, 通过对各个测量值按关联似然加权来生成单个状态假设。在 MHT 中, 所有可能的假设都会被跟踪, 但为了便于计算, 必须应用剪枝方案。这两种方法最近都在基于检测的跟踪场景 [10, 11] 中被重新审视, 并展现出良好的效果。然而, 这些方法的性能提升是以增加计算和实现复杂度为代价的。简单在线实时跟踪 (SORT) 12 是一个更为简单的框架, 它在图像空间中执行卡尔曼滤波, 并使用匈牙利方法进行逐帧数据关联, 关联度量用于测量边界框重叠度。这种简单的方法在高帧率下取得了良好的性能。在 MOT 挑战数据集 13 上, 配备最先进人体检测器 14 的 SORT 在标准检测中的平均排名高于 MHT。这不仅强调了物体检测器性能对整体跟踪结果的影响, 而且从实践者的角度来看也是一个重要的见解。虽然 SORT 在跟踪精度和准确度方面取得了良好的整体性能, 但它返回的身份切换次数相对较高。这是因为所采用的关联度量仅在状态估计不确定性较低时才准确。因此, SORT 在遮挡情况下的跟踪存在不足, 因为遮挡通常出现在正面视角的摄像头场景中。我们通过用结合运动和外观信息的更明智的度量来替换关联度量来解决这个问题。具体来说, 我们应用了一个卷积神经网络 (CNN), 该网络已经在大规模行人重识别数据集上训练用于识别行人。通过集成该网络, 我们提高了对漏检和遮挡的鲁棒性, 同时保持

了系统易于实现、高效且适用于在线场景。我们的代码和预训练的 CNN 模型已公开发布，以促进研究实验和实际应用开发以及可用于生成特征的本脚本。

实验 我们在 MOT16 基准 15 上评估了跟踪器的性能。该基准在七个具有挑战性的测试序列上评估跟踪性能，包括 带有移动摄像机的正面场景以及自上而下的监控场景。我们依赖 Yu 等人 16 提供的检测数据作为跟踪器的输入。他们在一系列公共和私有数据集上训练了一个 Faster RCNN，以提供出色的性能。为了进行公平的比较，我们对相同的检测数据重新运行了 SORT。使用 $\lambda = 0$ 和 $A_{max} = 30$ 帧对测试序列进行评估。与 16 中一样，检测的阈值设置为 0.3 的置信度得分。我们方法的其余参数是在基准提供的单独训练序列中找到的。评估依据以下指标进行：

- 多目标跟踪准确率 (MOTA)：根据误报、漏报和身份转换情况，对整体跟踪准确率进行总结 23。
- 多目标跟踪精度 (MOTP)：根据地面实况与报告位置之间的边界框重叠情况，对整体跟踪精度进行总结 23。
- 大部分跟踪 (MT)：在至少 80% 的生命周期内具有相同标签的地面实况轨迹所占百分比。
- 大部分丢失 (ML)：在最多 20% 的生命周期内被跟踪的地面实况轨迹所占百分比。
- 身份转换 (ID)：报告的地面实况轨迹身份发生变化的次数。
- 碎片化 (FM)：轨迹因漏检而中断的次数。

我们的评估结果如表 2 所示。我们的改进成功减少了身份切换的次数。与 SORT 相比，身份切换次数从 1423 次减少到 781 次，减少了约 45%。同时，由于需要通过遮挡和未命中来保持目标身份，轨迹碎片略有增加。我们还发现，大部分被跟踪的目标数量显著增加，而大部分丢失的目标数量有所减少。总体而言，由于整合了外观信息，我们成功地在较长的遮挡时间内保持了身份。这一点也可以通过我们在补充材料中提供的跟踪输出的定性分析来证明。图 1 显示了我们跟踪器的示例输出。我们的方法也是其他在线跟踪框架的有力竞争对手。特别是，我们的方法在所有在线方法中返回的身份切换次数最少，同时保持了具有竞争力的 MOTA 得分、轨迹碎片和误报率。报告的跟踪精度 主要受大量误报的影响。考虑到误报对 MOTA 得分的整体影响，对检测应用更大的置信度阈值可能会 大幅提升我们算法的报告性能。然而，目视检查跟踪输出表

明，这些误报大多是由静态场景几何形状下的零星检测器响应产生的。由于我们允许的最大轨迹年龄相对较大，这些误报通常与物体轨迹相结合。同时，我们没有观察到轨迹在误报之间频繁跳跃。相反，跟踪器通常在报告的物体位置生成相对稳定、静止的轨迹。我们的实现以大约 20 Hz 的频率运行，其中大约一半的时间用于特征生成。因此，在现代 GPU 的帮助下，该系统仍然保持计算效率并实时运行。

结论 我们提出了一种 SORT 扩展算法，通过预先训练的关联度量来整合外观信息。得益于此扩展算法，我们能够在较长时间的遮挡下进行跟踪，这使得 SORT 成为最先进的在线跟踪算法的有力竞争对手。同时，该算法仍然易于实现且实时运行。

安徽理工大学本科毕业论文（设计）开题报告

姓 名	陈亚辉	专业班级	计算机科学与技术计算机21-4	指导教师	郑苹
一、课题的名称、来源					
1.课题名称	基于深度学习的车道障碍检测及碰撞预防方法研究				
2.课题来源	☐ 生产 ☒ 科研 ☐ 教学 ☐ 其他				
二、研究意义、研究现状、研究内容、拟采用的研究思路与方法(可附页)					
1. 研究意义					
实践意义					
交通安全提升：我国汽车保有量已达 3.36 亿辆，驾驶员数量达 4.86 亿人，巨大的市场规模伴随显著的安全挑战。全球每年约有 135 万人死于道路交通事故，致伤 2000 万至 5000 万人，其中 94%的事故与人为因素直接相关。					
经济损失减少：交通事故造成的经济损失约占多数国家 GDP 的 3%，通过智能驾驶辅助技术可有效降低这一损失。					
自动驾驶技术支持：研究表明自动驾驶车辆在多数场景下的安全性优于人工驾驶，其追尾和侧面碰撞风险分别仅为人工驾驶的 45.7%和 17.1%，特别是在车道保持和自适应车距调节方面表现突出。					
理论意义					
弥补传统驾驶的三重安全隐患：感知局限、反应迟滞和行为不可控，这些问题是导致大多数交通事故的关键因素。					
推进智能驾驶技术发展：通过结合计算机视觉和深度学习算法，为自动驾驶和辅助驾驶系统提供可靠的障碍物检测和碰撞预测技术支持。					
促进跨学科融合：研究整合了图像处理、深度学习、目标跟踪和风险评估等多学科技术，推动相关领域的交叉融合创新。					
2. 研究现状					
车道线检测技术发展					
传统方法：早期车道线检测主要依赖颜色、边缘等基础图像特征。这些方法虽然速度较快，但在复杂场景下(车辆密集、雨雪天气或光线不足)表现欠佳，难以满足辅助驾驶系统的实际需求。					
深度学习方法：近年来，深度神经网络和大规模道路数据集的应用为道路分割提供了新的解决方案。如 SCNN 采用逐切片的空间卷积策略，增强特征图行列间的像素信息交流，有效提高了细长结构的识别能力。					
优化策略：为提升推理速度，研究者开发了多种优化方法，包括剪枝、量化、知识蒸馏和轻量级分割网络等，以适应实时系统的需求。					
障碍物检测与跟踪技术					
检测方法分类：目前障碍物检测主要分为接触式和非接触式两类。接触式检测主要基于传感器技术，非接触式检测则主要依靠机器视觉技术。					

国内研究进展：国内研究主要包括 ViBe 算法的运动目标检测，以及将 ViBe 与光流法、帧差法及深度学习相结合的新型目标检测算法。

多目标跟踪：在高速公路车辆多目标跟踪领域，多特征融合已成为提高跟踪性能的重要研究方向。基于深度特征和颜色直方图融合的车辆多目标跟踪算法，通过综合利用深度学习提取的语义特征和传统视觉特征，在复杂场景下展现出良好的鲁棒性和稳定性。

3. 研究内容

本研究主要围绕基于深度学习的车道障碍检测与碰撞预测方法展开，具体包含以下几个方面：

车道规划

车道线提取：通过颜色分割、多尺度 DoG 边缘检测、边缘增强、区域限定、直线检测和车道线拟合等步骤，实现对车道线的精确提取。

FastSCNN 语义分割：选取适合的数据集，进行预处理、模型训练和评估，利用深度学习实现道路区域的语义分割。

图像处理与语义分割融合：结合传统图像处理和深度学习的语义分割方法，提高车道识别的准确性和鲁棒性。

障碍物监测

障碍检测：包括数据集预处理、障碍目标检测、模型训练与评估以及模型蒸馏及量化等环节，实现对道路障碍物的快速准确识别。

障碍跟踪：通过改进的 DeepSORT 框架、卡尔曼滤波器、匈牙利算法级联匹配和特征度量等技术，实现对多目标的实时跟踪。

碰撞预测

运动轨迹预测：基于目标当前和历史状态，采用适合嵌入式系统的线性预测模型，推断目标未来可能的位置。

碰撞时间计算：建立 TTC(Time To Collision)计算模型，评估风险紧迫性，为及时预警提供依据。

综合风险评估：结合多种因素，建立综合风险评估模型，实现碰撞风险的精确预测和分级预警。

4. 拟采用的研究思路与方法

整体技术框架

研究采用分层结构设计，将车道障碍检测与碰撞预测问题拆解为多个子模块，包括视频图像序列采集、道路特征提取与语义分割、基于深度学习的目标检测、复杂场景多目标跟踪、碰撞概率建模与分级预警等，最终实现一个完整的障碍检测与碰撞预测系统。

关键技术方法

车道规划技术

1. 传统图像处理方法：

颜色分割：利用车道线的颜色特征进行初步筛选

多尺度 DoG 边缘检测：增强对不同尺度边缘的识别能力

边缘增强和区域限定：提高车道线特征的突出度，减少干扰

2. 深度学习语义分割：

采用 FastSCNN 轻量级模型降低计算负担

多尺度特征融合：提高对不同大小目标的识别能力

传统方法与深度学习协同机制：结合两者优势，提高识别准确性和鲁棒性

障碍物检测与跟踪技术

1. 目标检测：

采用 YOLOv5 目标检测算法，实现对车辆、行人等障碍物的快速识别

模型蒸馏及量化：减少计算资源消耗，提高实时性

2. 多目标跟踪：

改进 DeepSORT 算法：结合深度特征和传统视觉特征

卡尔曼滤波器：预测目标运动状态

匈牙利算法级联匹配：实现目标 ID 的稳定分配

特征度量优化：提高跟踪精度和稳定性

碰撞预测技术

1. 运动轨迹预测：

线性预测模型：基于目标当前位置和速度向量进行短期预测

适配计算资源受限的嵌入式系统，平衡预测准确性和计算效率

2. 碰撞时间计算：

TTC 基本计算模型：基于两物体间距离和相对速度

区域碰撞检测方法：考虑物体实际形状和尺寸

动态 TTC 阈值调整：根据车速、道路类型和环境条件进行优化

3. 风险评估模型：

三级预警机制：根据碰撞风险程度进行分级预警

综合各类因素的多维风险评估，提供更精确的危险预测

优化与部署

模型优化：通过模型剪枝、量化和知识蒸馏等技术，降低模型计算资源需求

低成本部署：针对车载设备的资源限制进行系统优化，使其能够在低成本设备上高效运行

实时性保证：通过算法优化和并行处理提高系统响应速度，确保及时预警

通过上述研究思路与方法的综合应用，本研究旨在构建一个准确、实时、鲁棒的车道障碍检测与碰撞预测系统，为提高道路交通安全提供技术支持

三、主要参考文献

- [1] 中国汽车工业协会. 2023 年汽车工业经济运行报告 [EB/OL]. 2023 [2025-04-30]. <http://lwzb.stats.gov.cn/pub/lwzb/bztt/202405/W020240527578179613545.pdf>.
- [2] World Health Organization. Global status report on road safety 2023[EB/OL]. World Health Organization. 2023 [2025-04-30]. <https://www.who.int/teams/social-determinants-of-health/safety-and-mobility/global-status-report-on-road-safety-2023>.
- [3] Abdek-Aty M, Ding S. A matched case-control analysis of autonomous vs human-driven vehicles accidents[J]. Nature Communications, 2024, 15(1): 4931-4942.
- [4] Schwall M, Daniels T, Victor T, et al. Waymo public road safety performance data[J]. arXiv preprint, 2020. arXiv: 2011.00038.
- [5] Asgari Taghanaki S, Abhishek K, Cohen J P, et al. Deep semantic segmentation of natural and medical images: a review[J]. Artificial intelligence review, 2021, 54: 137-178.
- [6] Parashar A, Rhu M, Mukkara A, et al. SCNN: An accelerator for compressed-sparse convolutional neural networks[J]. ACM SIGARCH computer architecture news, 2017, 45(2): 27-40.
- [7] Zhu M, Gupta S. To prune, or not to prune: exploring the efficacy of pruning for model compression [J]. arXiv preprint arXiv:1710.01878, 2017.
- [8] Park J, No A. Prune your model before distill it[C]//European Conference on Computer Vision. 2022:120-136.
- [9] Huang Q, Zhou K, You S, et al. Learning to prune filters in convolutional neural networks[C]//2018 IEEE Winter Conference on Applications of Computer Vision (WACV). 2018: 709-718.
- [10] Goodman D, Wilkinson R. A robust adaptive quantizer[J]. IEEE Transactions on Communications, 1975, 23(11): 1362-1365.
- [11] Fischer T. A pyramid vector quantizer[J]. IEEE transactions on information theory, 1986, 32(4): 568-583.
- [12] An S, Shin J, Kim J. Quantization-aware training with Dynamic and Static Pruning[J]. IEEE Access, 2025.
- [13] Hou Y, Ma Z, Liu C, et al. Learning lightweight lane detection cnns by self attention distillation[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2019: 1013-1021.
- [14] Hou Y, Ma Z, Liu C, et al. Inter-region affinity distillation for road marking segmentation[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020: 12486-12495.
- [15] Park W, Kim D, Lu Y, et al. Relational knowledge distillation[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2019: 3967-3976.
- [16] Paszke A, Chaurasia A, Kim S, et al. Enet: A deep neural network architecture for real-time semantic segmentation[J]. arXiv preprint arXiv:1606.02147, 2016.
- [17] 刘彬辉. 基于结构和稀疏特性的车道线检测算法研究[D]. 中国科学技术大学, 2024.
- [18] 陈文艺. 基于深度学习的城市街景语义分割方法研究[D]. 西京学院, 2024.
- [19] Discant A, Rogozan A, Rusu C, et al. Sensors for obstacle detection-a survey[C]//2007 30th International Spring Seminar on Electronics Technology (ISSE). 2007: 100-105.
- [20] Badrloo S, Varshosaz M, Pirasteh S, et al. Image-based obstacle detection methods for the safe navigation of unmanned vehicles: A review[J]. Remote Sensing, 2022, 14(15): 3824.
- [21] Barnich O, Van Droogenbroeck M. ViBe: A universal background subtraction algorithm for videosequences[J]. IEEE Transactions on Image processing, 2010, 20(6): 1709-1724.
- [22] 费莉梅, 田翔, 郑博仑. 基于改进 ViBe 的自适应运动目标检测算法[J]. 计算机工程与设计,

2024,45(06): 1771-1779.

[23] 张颜月, 代显智. 复杂环境下运动目标检测研究进展[J]. 四川轻化工大学学报 (自然科学版), 1-9.

[24] 张雷, 王延杰, 孙宏海, 等. 基于颜色和边缘直方图的多目标跟踪[J]. Chinese Journal of Liquid Crystal & Displays, 2016, 31(6).

[25] Wang Q, Cao L, Xia J, et al. Mtcnn-kcf-deepsort: Driver face detection and tracking algorithm based on cascaded kernel correlation filtering and deep sort[R/OL]. SAE Technical Paper. 2020 [2025-04-30]. <https://www.sae.org/publications/technical-papers/content/2020-01-1038/>.

[26] Kong H, Akakin H C, Sarma S E. A generalized Laplacian of Gaussian filter for blob detection and its applications[J]. IEEE transactions on cybernetics, 2013, 43(6): 1719-1733.

[27] Chen Z, Zhou Z, Adnan S. Joint low-rank prior and difference of Gaussian filter for magnetic resonance image denoising[J]. Medical & Biological Engineering & Computing, 2021, 59: 607-620.

[28] Xie S, Tu Z. Holistically-nested edge detection[C]//Proceedings of the IEEE international conference on computer vision. 2015: 1395-1403.

[29] Liu Y, Cheng M M, Hu X, et al. Richer convolutional features for edge detection[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2017: 3000-3009.

[30] Illingworth J, Kittler J. A survey of the Hough transform[J]. Computer vision, graphics, and image processing, 1988, 44(1): 87-116.

[31] Poudel R P, Liwicki S, Cipolla R. Fast-scnn: Fast semantic segmentation network[J]. arXiv preprint arXiv:1902.04502, 2019.

[32] Dalal N, Triggs B. Histograms of oriented gradients for human detection[C]//2005 IEEE computer society conference on computer vision and pattern recognition (CVPR' 05): vol. 1. 2005: 886-893.

[33] Lowe D G. Distinctive image features from scale-invariant keypoints[J]. International journal of computer vision, 2004, 60: 91-110.

[34] Wojke N, Bewley A, Paulus D. Simple online and realtime tracking with a deep association metric[C]//2017 IEEE International Conference on Image Processing (ICIP). 2017: 3645-3649.

四、进度安排

2025 年 02 月 17 日-2025 年 03 月 14 日

学习 YOLOv5 模型结构与原理

研究模型轻量化与优化方法

确定目标检测分类体系(行人、动物、非机动车等)

设计碰撞风险评估算法

收集或制作道路场景数据集

数据标注与预处理

设计模型训练策略

开始 YOLOv5 模型训练与优化

五、指导教师意见

同意

指导教师签名：郑苹		日期：2025 年 05 月 23 日
六、所在系意见 同意指导教师意见		
负责人签名（签章）：蒋社德		日期：2025 年 05 月 29 日

注：此件由学生本人填写，一式二份，院（部）、指导教师各存一份。

安徽理工大学本科毕业论文（设计）指导过程记录表

题 目	基于深度学习的车道障碍检测及碰撞预防方法研究				
学生姓名	陈亚辉	学号	2021304369	专业班级	计算机科学与技术计算机 21-4
指导教师	郑苹	职称	讲师	教研室	数字媒体技术
指导内容 记录(一)	指导记录 1 在第一次指导中，郑老师详细介绍了毕业论文的整体流程与重要性，并针对计算机视觉与深度学习领域展开讨论。老师耐心倾听了我提出的几个潜在选题，并对每个方向的可行性和学术价值进行了分析。针对资料收集方面，郑老师推荐了几个权威学术网站和数据库，如 IEEE Xplore、中国知网等，并指导我如何筛选和整理有价值的文献。同时，我们共同制定了详细的论文写作计划表，明确了各阶段的目标和时间节点。 时间：2025 年 02 月 17 日				
指导内容 记录(二)	指导记录 2 经过深入探讨，我最终确定了论文题目《基于深度学习的车道障碍检测与碰撞预防方法研究》。郑老师引导我分析了该选题的学术价值、应用前景以及与当前智能驾驶技术发展的契合点。针对论文写作，老师详细讲解了学术论文的结构规范、引用标准以及学术表达的技巧。特别是在文献收集方面，郑老师建议我从经典算法到最新研究成果进行全面梳理。在老师的指导下，我完成了文献综述初稿，并确定了论文的整体框架和研究方法。 时间：2025 年 02 月 14 日				
指导内容 记录(三)	指导记录 3 按照郑老师提供的开题报告模板，我完成了开题报告的撰写工作。报告包括研究背景、目的意义、国内外研究现状、研究内容和技术路线等部分。随后，我着手撰写论文的"引言"和"背景"章节初稿。郑老师对初稿进行了细致点评，指出了论述逻辑和学术表达需要改进的地方。她建议我更加突出智能驾驶技术的发展趋势和挑战，并补充更多关于计算机视觉在道路场景理解中的应用案例，使背景介绍更具说服力。 时间：2025 年 03 月 03 日				
指导内容	指导记录 4 在郑老师的建议下，我深入研究了国内外计算机视觉领域的最新成果，特别是图像处理和目标跟踪相关技术。通过大量阅读论文和技术报告，我系统梳理了从传统图像处				

记录(四)	理到深度学习的技术演进路径。同时，我下载并学习了多个开源平台的代码实现，包括语义分割框架和各代 YOLO 目标检测器。在实践层面，我使用 PyTorch 框架搭建了测试环境，对多种模型进行了训练和评估，深入理解了端到端目标检测和语义分割的核心原理与实现细节。 时间：2025 年 03 月 10 日
指导内容 记录(五)	指导记录 5 完成资料整理后，我与郑老师讨论了系统整体架构和技术路线。我们确定了采用模块化设计思路，将车道线识别、障碍物检测和碰撞风险评估作为三个核心功能模块。针对每个模块，我们详细讨论了算法选型、数据流和接口设计。郑老师建议我结合轻量级网络和模型压缩技术，以满足实时性要求。根据讨论结果，我完成了系统架构初稿，明确了技术细节和实现路径，为后续开发奠定了基础。 时间：2025 年 03 月 17 日
指导内容 记录(六)	指导记录 6 这阶段重点是数据准备和模型选型。我下载并预处理了多个道路场景数据集，包括 BDD100K 和 Cityscapes 等，对图像进行了标准化和增强处理，使其符合深度学习模型的输入要求。郑老师指导我绘制了系统的技术流程图和各模块的算法流程图，使整个系统设计更加清晰可视。同时，我准备了实验所需的硬件设备，包括 GPU 工作站和边缘计算设备，为后续实验测试做好了准备。我们还讨论了不同模型在精度和速度上的权衡问题。 时间：2025 年 03 月 24 日
指导内容 记录(七)	指导记录 7 在郑老师指导下，我专注于模块化代码实现和系统优化。针对内存受限的边缘设备，我引入了轻量化库和优化的算法实现。郑老师与我深入讨论了实时系统设计原则和硬件接口问题，帮助我理解了多线程编程的重要性。我实现了线程池管理机制，有效提升了 CPU 利用率和系统响应速度。通过反复测试，成功验证了各模块接口的兼容性和稳定性。同时，我完成了开发环境、训练环境和测试环境的配置，为后续实验奠定了基础。 时间：2025 年 03 月 31 日
指导内容 记录(八)	指导记录 8 此阶段重点是模型训练和优化。我完成了语义分割网络和两个 YOLOv5 变种模型（small 和 nano 版本）的训练工作，并详细记录了训练过程和评估指标。郑老师和我探讨了多种模型压缩技术的应用场景和效果，最终确定采用知识蒸馏和静态量化两种方法来平衡模型性能和资源消耗。知识蒸馏使小模型能够学习大模型的特征表达能力，而静态量化则显著减少了模型大小和推理时间，使系统更适合在资源受限设备上部署。

	时间：2025 年 04 月 07 日
指导内容 记录(九)	指导记录 9 在郑老师的耐心指导下，我完成了系统各模块的整合和细节优化。针对车道线识别、障碍物检测和碰撞预警三个模块，我实现了数据流转换和结果融合算法。将系统烧录到测试硬件后，我们发现了几个性能瓶颈和逻辑错误，我与郑老师一起分析了问题根源并进行了针对性修正。特别是在处理复杂光照条件和遮挡情况时，我们调整了检测阈值和后处理策略，显著提升了系统鲁棒性。郑老师还指导我优化了整体技术流程，减少了冗余计算。 时间：2025 年 04 月 14 日
指导内容 记录(十)	指导记录 10 项目完成后，郑老师与我进行了系统的部署前景讨论。我们分析了该系统在智能驾驶辅助、智慧交通和安全监控等领域的应用潜力，并探讨了商业化推广策略。郑老师帮助我梳理了系统的经济效益分析，包括开发成本、维护成本与可能带来的安全价值提升。我们还列出了未来可改进的方向，如引入注意力机制、多模态融合和强化学习等技术。最后，我们确定了论文的创新点，包括轻量化模型设计和实时碰撞风险评估算法，为后续理论研究打下基础。 时间：2025 年 04 月 21 日
指导内容 记录(十一)	指导记录 11 在前期充分准备的基础上，我用一周时间完成了论文初稿。郑老师对初稿进行了详细审查，从理论框架、研究方法、数据分析到结论推导各个方面都给出了宝贵意见。特别是在创新点阐述部分，郑老师建议我加强与现有方法的对比分析，更清晰地展示本研究的优势和贡献。在实验结果展示方面，她指导我增加了更多定量分析和可视化图表，使论文更具说服力和学术价值。我们还讨论了论文的实用价值和局限性，使研究更加全面和客观。 时间：2025 年 04 月 28 日
指导内容 记录(十二)	指导记录 12 根据郑老师的修改建议，我全面修订了论文内容。首先，我重新梳理了理论框架，加强了各章节之间的逻辑衔接；其次，我规范了学术表达，修正了不准确的术语使用；最后，我优化了图表和公式的排版，提升了论文的专业性和可读性。在技术细节描述方面，我增加了算法伪代码和流程图，使实现过程更加清晰。我还广泛征求了同学和其他老师的意见，根据反馈进一步完善了论文质量。

	时间：2025 年 05 月 06 日
指导内容 记录(十三)	指导记录 13 这阶段重点是论文的最终校对和查重。我完成了参考文献的格式规范和内容核对，确保引用准确无误。郑老师指导我使用维普系统进行论文查重测试，初步结果显示查重率在可接受范围内。针对查重报告中标注的部分内容，我进行了适当修改，包括改写表达方式、增加自己的分析见解等。郑老师特别强调了学术创新点的表述要准确恰当，避免过度宣称或表述不清的问题，确保论文具有较高的学术水准。 时间：2025 年 05 月 12 日
指导内容 记录(十四)	指导记录 14 临近答辩，郑老师与我一起进行了论文的最终审查。我们逐章节检查了内容的准确性和完整性，特别是技术细节和实验数据的准确性。郑老师指导我准备了答辩 PPT，建议我突出研究的创新点、关键技术和实验结果。她分享了答辩技巧，包括如何把握时间、如何清晰表达核心观点以及如何应对提问。我们还模拟了可能的答辩问题，如算法选择依据、实验设计合理性以及系统局限性等，使我能够更加从容地面对答辩。 时间：2025 年 05 月 19 日
指导内容 记录(十五)	指导记录 15 完成所有准备工作后，我将论文终稿和附件材料提交给郑老师进行最后审核。同时，我精心准备了答辩演讲稿，反复练习以确保在规定时间内清晰表达研究内容和成果。针对可能出现的质疑，我准备了详细的解答思路和佐证材料。郑老师对我的演讲进行了指导，建议我放慢语速、强调重点内容，并准备了一些辅助说明的图表。通过这些准备，我对即将到来的答辩充满信心，相信能够出色地展示我的研究成果。 时间：2025 年 05 月 23 日

注：1）此表格是学生毕业设计（论文）归档材料之一，由学生填写，一周填一格，手写、打印均可；
2）表格内容不够填写可另附纸。

安徽理工大学本科毕业论文（设计）中期检查表

学号	2021304369	姓名	陈亚辉	专业班级	计算机科学与技术计算机 21-4
指导教师	郑苹	检查人	郑苹	检查日期	2025 年 04 月 18 日
题目	基于深度学习的车道障碍检测及碰撞预防方法研究				
设计（论文）进度计划	1/2				
目前已经完成内容	1. 绪论部分： 研究背景及意义的撰写 国内外研究进展的文献调研与总结 论文整体架构的规划与确定 2. 相关理论基础： 图像处理基础理论整理，包括传统图像增强技术、形态学运算、边缘检测和车道线拟合 语义分割技术概述，重点阐述 FastSCNN 模型和多尺度特征融合策略 目标检测与目标跟踪的理论框架，特别是 YOLOv5 和 DeepSORT 算法的机制分析 3. 部分方法设计与实验准备： 技术路线图的设计 实验数据集的选取与预处理 基本实验环境的搭建				
尚需完成的内容	1. 车道规划章节（第三章）： 车道线提取算法的具体实现与优化 FastSCNN 模型的训练流程与参数调优 图像处理与语义分割融合机制的设计与验证 相关实验结果分析与展示 2. 障碍监测章节（第四章）： 障碍检测模块的详细实现 模型蒸馏及量化方案的制定与实施 障碍跟踪算法的改进与实现 多目标跟踪性能的定量评估与分析 3. 碰撞预测章节（第五章）： 运动轨迹预测算法的实现与测试 碰撞时间计算模型的构建 综合风险评估模型的建立与验证 预警机制的设计与实现 4. 总结与展望（第六章）： 研究成果的系统总结				

	研究不足的分析与反思 未来研究方向的展望 5. 论文完善工作： 插图、表格的制作与完善 参考文献的规范化整理 论文格式的统一与校对
存在的问题和拟采取的办法	1. 算法实时性问题： 问题：深度学习模型计算资源需求大，可能影响系统实时性 解决方法： 采用模型剪枝、量化和知识蒸馏技术降低计算复杂度 设计并行处理流程，优化算法实现 针对关键模块进行代码优化，提高执行效率 2. 实验数据获取难题： 问题：不同道路环境和天气条件下的测试数据获取困难 解决方法： 结合公开数据集与自采数据，构建多样化测试集 使用数据增强技术扩充样本多样性 必要时采用仿真环境进行补充测试 3. 算法鲁棒性挑战： 问题：在复杂环境下（如雨雪天气、光照变化）算法性能可能下降 解决方法： 增加极端条件下的训练样本 设计自适应阈值调整机制 结合多模态信息提高系统稳定性 4. 评估指标的确定： 问题：缺乏统一的评估标准来衡量系统整体性能 解决方法： 参考相关领域的标准评估指标 设计针对特定应用场景的自定义评估指标 进行多角度、多维度的综合评估 5. 理论与实践结合问题： 问题：理论模型与实际应用场景可能存在差距 解决方法： 进行真实环境测试验证 针对实际问题进行算法调整 结合专家经验对系统进行优化
指导教师 签名	

安徽理工大学

本科毕业论文（设计）评阅人评阅表

专业、班级 计算机科学与技术计算机 21-4 姓名 陈亚辉 完

成日期 2025 年 05 月 20 日

论文（设计）题目 基于深度学习的车道障碍检测及碰撞预防方法研究

评分维度	评分标准	分值	得分
1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符合的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	9
2. 论证、分析、计算	方案论证充分、研究方法合理；综合分析的正确、合理性；设计、计算的正确性。	20	17
3. 结构、建模、实验	结构的正确、合理性；数模建立的正确、合理性；实验合理性、数据可靠性。	20	17
4. 说明书（或论文）质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计（或论文）的实用性与科学性。	25	23
5. 图纸（或插图）质量、各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	13
6. 创新性	工作中有创新意识或有独特见解。	10	7
总得分			86

评阅人意见：[对论文（设计）优缺点、修改建议等]

题目与专业人才培养目标相符合；难易程度程度，达到了综合训练的目的；具有一定的独立查阅资料、进行调研的能力；有收集处理相关信息及获得新知识的能力。方案论证充分、研究方法合理；综合分析合理；设计正确。结构正确、数模建立较合理；实验数据可靠。条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计的实用性较好。布局合理、表达清晰；图面质量符合要求；附件材料齐全、内容完整。工作中有创新意识。

是否同意答辩：☒同意 ☐修改后答辩 ☐不同意

评阅人：

日期：2025 年 05 月 27 日

安徽理工大学

本科毕业论文（设计）指导教师评阅表

专业、班级 计算机科学与技术计算机 21-4 姓名 陈亚辉 完

成日期 2025 年 05 月 20 日

论文（设计）题目 基于深度学习的车道障碍检测及碰撞预防方法研究

评分维度	评分标准	分值	得分
1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符合的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	9
2. 论证、分析、计算	方案论证充分、研究方法合理；综合分析的正确、合理性；设计、计算的正确性。	15	14
3. 结构、建模、实验	结构的正确、合理性；数模建立的正确、合理性；实验合理性、数据可靠性。	15	14
4. 说明书（或论文）质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计（或论文）的实用性与科学性。	20	17
5. 图纸（或插图）质量、各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	14
6. 创新性	工作中有创新意识或有独特见解。	10	9
7. 工作态度与进度	按时完成任务，主动沟通，态度认真，尊重学术规范。	10	9
8. 外文文献翻译	外文文献翻译评分表得分×5%。	5	4.30
总得分			90.30

是否同意答辩：☒同意 ☐不同意

指导教师： 郑苹 日期： 2025 年 05 月 26 日

安 徽 理 工 大 学

本科毕业论文（设计）答辩成绩评定表

专业、班级 计算机科学与技术计算机 21-4 姓名 陈亚辉 完

成日期 2025 年 05 月 20 日

论文（设计）题目 基于深度学习的车道障碍检测及碰撞预
防方法研究

答辩地点： 求本楼 A311 答辩日期： 2025 年 05 月 27 日

答辩提问录

序号	提问人	提问主要问题	对答辩人回答问题的评语
1	许光宇	Q: 你的论文主要研究的方向是什么，对什么进行处理、怎么演示 A: 我的论文主要是对智能辅助驾驶系统的研究，通过车道规划、障碍监测、碰撞预测三个模块对摄像头采集的帧序列进行处理。这里使用 Python 对视频处理的过程进行展示效果。	

2	张玉	Q: 你的采用的什么方式对车道划分? A: 通过传统的图像处理算法和语义分割模型 FastSCNN 联合, 采用这种“白”+“黑”的模式, 可以提高整个车道划分的健壮性。	
3	郑苹	Q: 你的文中提到的障碍主要是指的是什么 A: 障碍主要包括道路上的车辆、行人、自行车等一类的活动障碍, 以及路牌、交通信号标志等静态障碍物。通过使用 YOLOv5 作为目标检测器, 在使用 DeepSORT 算法对检测到的目标跟踪从而实现对道路上的各种障碍的监测。	
4	许光宇	Q: 为什么检测背景使用一个绿色的蒙版 A: 因为对于大部分车载场景, 摄像采集的画面的上方 30%~45% 是一些天空、高楼等非重要的信息, 而通过一个 ROI 区域	

		限定,可以更专注于道路上的目标。	
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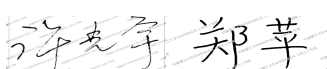
答辩成绩

评分维度	评分标准	分值	答辩小组 平均得分
1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符合的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	8
2. 说明书（或论文）质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计（或论文）的实用性与科学性。	15	13
3. 图纸（或插图）质量、各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	13
4. 创新性	工作中有创新意识或有独特见解。	15	13
5. 内容陈述	PPT 清晰（如有），重点突出，语言流畅，时间控制合理。	15	13
6. 问题回答	回答准确，逻辑性强，能有效辩护论文观点。	20	18
7. 答辩表现	仪态端正，表达自信，与评委互动良好。	10	8
总得分			86

答辩小组意见：[对论文（设计）优缺点、修改建议等]

题目与专业人才培养目标相符；难易适度，较好地达到了毕业设计综合训练的目的；独立查阅资料、进行调研的能力较强；有一定的收集处理相关信息及获得新知识的能力。条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计的实用性较好。布局合理、表达清晰；图面质量符合要求；附件材料齐全、内容完整。工作中创新意识较好。答辩期间语言较流畅，时间控制合理。回答基本正确，逻辑性较好，能有效辩护论文观点。仪态端正，表达非常自信，与评委互动较好。

答辩秘书（签名）：

答辩小组（签名）：

日期：2025 年 05 月 28 日

安徽理工大学

本科毕业论文（设计）总评成绩表

专业、班级 计算机科学与技术计算机 21-4 姓名 陈亚辉 完

成日期 2025 年 05 月 20 日

毕业论文（设计）题目 基于深度学习的车道障碍检测及碰撞预防方法研究

1. 毕业论文（设计）成绩的评定

指导教师（分）	评阅教师（分）	毕业答辩（分）	总评成绩（分）
90.30	86	86	87

注意：毕业论文（设计）总评成绩中指导教师的评分占 30%，评阅人评分占 30%，毕业答辩评分占 40%。

2. 专业答辩委员会

姓名	蒋社想	葛斌	方贤进	许家昌	王丽	刘唤唤	夏晨星	周华平
职称	副教授	教授	教授	副教授	讲师	讲师	副教授	教授
姓名	许欢							
职称	实验师							

专业答辩委员会主席（签名） 

日期： 2025 年 05 月 28 日